MEASUREMENTS & TECHNIQUES

1.3 Errors and uncertainties

Candidates should be able to:

1. understand and explain the effects of systematic errors (including zero errors) and random errors in measurements
2. understand the distinction between precision and accuracy
3. assess the uncertainty in a derived quantity by simple addition of absolute or percentage uncertainties

1 **9702/1/O/N/02/Q5**

A student carries out a series of determinations of the acceleration of free fall g. The table shows the results.

g/m s^{-2}
4.91
4.89
4.88
4.90
4.93
4.92

What can be said about this experiment?

A It is accurate and precise.

B It is accurate but not precise.

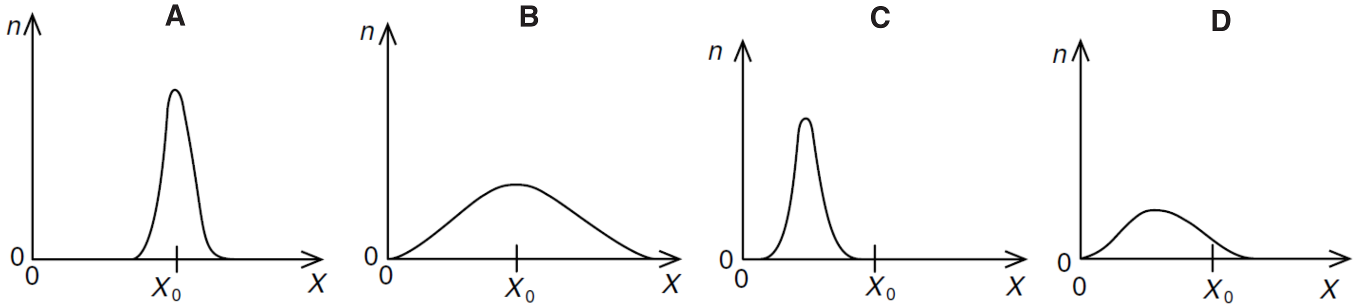
C It is not accurate and not precise.

D It is not accurate but is precise.

2 **9702/1/O/N/02/Q6**

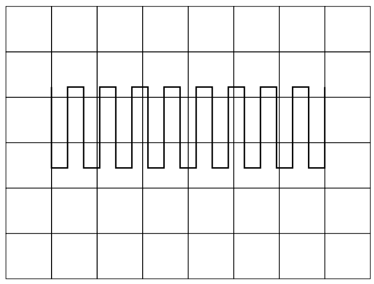
A quantity X is measured many times. A graph is plotted showing the number n of times a particular value of X is obtained. X has a true value X_0.

Which graph could be obtained if the measurement of X has a large systematic error but a small random error?



3 **9702/1/O/N/02/Q7***

The diagram shows a square-wave trace on the screen of a cathode-ray oscilloscope. A grid of 1 cm squares covers the screen. The time-base setting is 10 ms cm^{-1}.



What is the approximate frequency of the square-wave?

A 70 Hz **B** 140 Hz

C 280 Hz **D** 1400 Hz

4 **9702/01/O/N/03/Q4***

A thermometer can be read to an accuracy of ±0.5 °C. This thermometer is used to measure a temperature rise from 40 °C to 100 °C.

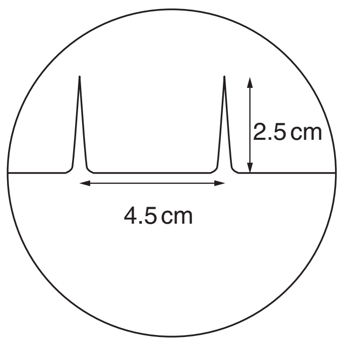
What is the percentage uncertainty in the measurement of the temperature rise?

A 0.5 % **B** 0.8 % **C** 1.3 % **D** 1.7 %

5 **9702/01/O/N/03/Q5**

The time-base on a cathode-ray oscilloscope is set at 6 ms / cm.

A trace consisting of two pulses is recorded as shown in the diagram.



What is the time interval between the two pulses?

A 0.42 ms **B** 0.75 ms

C 1.33 ms **D** 27 ms

6 9702/01/O/N/03/Q6

A micrometer screw gauge is used to measure the diameter of a copper wire.

The reading with the wire in position is shown in diagram 1. The wire is removed and the jaws of the micrometer are closed. The new reading is shown in diagram 2.

What is the diameter of the wire?

A 1.90 mm **B** 2.45 mm

C 2.59 mm **D** 2.73 mm

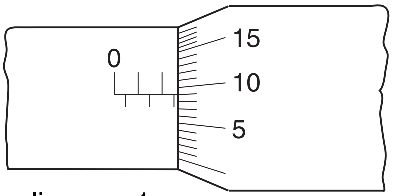


diagram 1

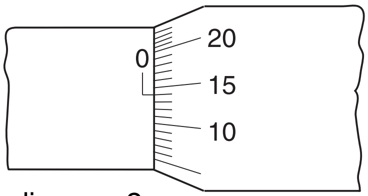


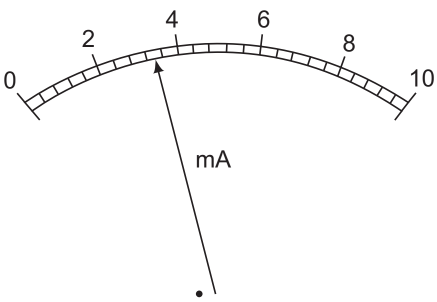
diagram 2

7 9702/01/M/J/04/Q4

What is the reading shown on this milliammeter?

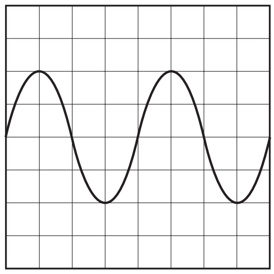
A 2.35 mA **B** 2.7 mA

C 3.4 mA **D** 3.7 mA



8 9702/13/M/J/16/Q5

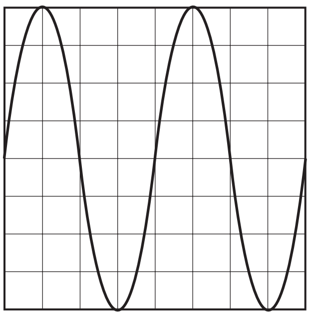
The following trace is seen on the screen of a cathode-ray oscilloscope.



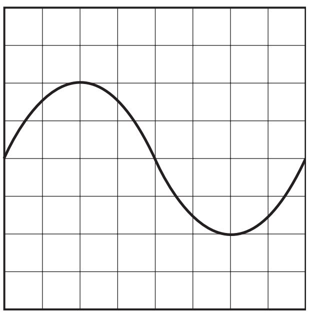
The setting of the time base is then changed from 10 ms cm^{-1} to 20 ms cm^{-1} and the Y-sensitivity is unaltered.

Which trace is now seen on the screen?

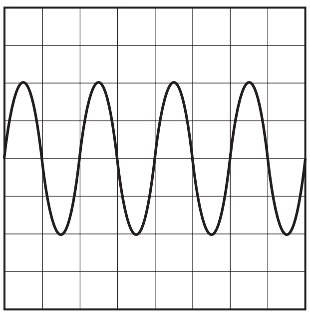
A



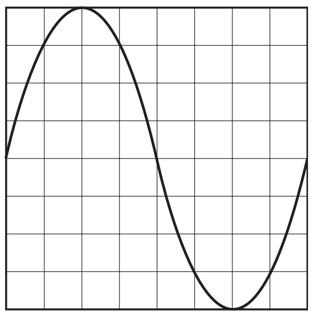
B



C



D



9 9702/13/M/J/16/Q5

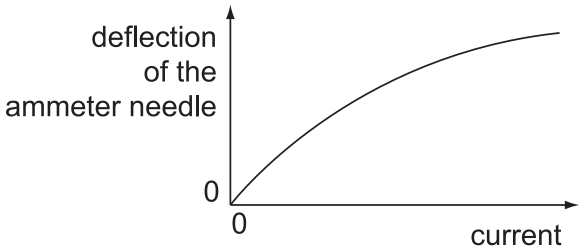
In a simple electrical circuit, the current in a resistor is measured as (2.50 ± 0.05) mA. The resistor is marked as having a value of 4.7 Ω ± 2 %.

If these values were used to calculate the power dissipated in the resistor, what would be the percentage uncertainty in the value obtained?

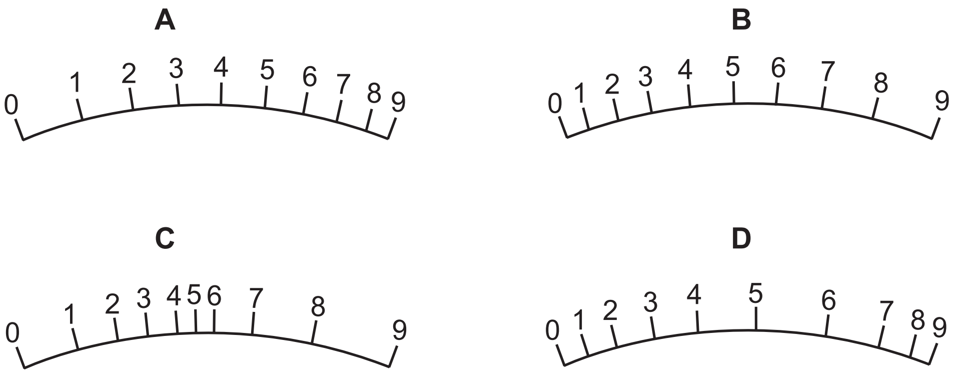
A 2 % **B** 4 % **C** 6 % **D** 8 %

10 **9702/01/O/N/04/Q4**

The deflection of the needle of an ammeter varies with the current passing through the ammeter as shown in the graph.

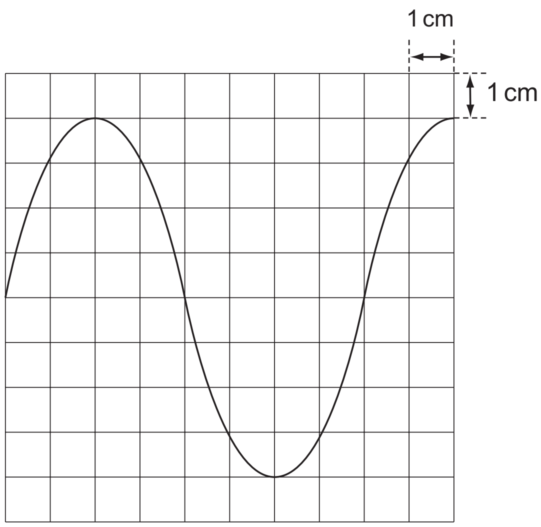


Which diagram could represent the appearance of the scale of this meter?



11 **9702/01/O/N/04/Q5**

When a 12 V 50 Hz supply is connected to the Y-terminals of an oscilloscope, the trace in the diagram is obtained.



What is the setting of the time-base control?

A $2.0\,\mathrm{ms\,cm^{-1}}$ **B** $2.5\,\mathrm{ms\,cm^{-1}}$

C $5\,\mathrm{ms\,cm^{-1}}$ **D** $20\,\mathrm{ms\,cm^{-1}}$

12 **9702/01/O/N/04/Q6**

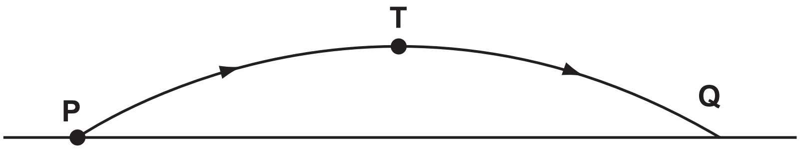
Four students each made a series of measurements of the acceleration of free fall g. The table shows the results obtained.

Which student obtained a set of results that could be described as precise but not accurate?

student	results, g / m s^{-2}			
A	9.81	9.79	9.84	9.83
B	9.81	10.12	9.89	8.94
C	9.45	9.21	8.99	8.76
D	8.45	8.46	8.50	8.41

13 9702/01/O/N/04/Q7

In the absence of air resistance, a stone is thrown from **P** and follows a parabolic path in which the highest point reached is **T**. The stone reaches point **Q** just before landing.



The vertical component of acceleration of the stone is

A zero at **T**. **B** greatest at **T**. **C** greatest at **Q**. **D** the same at **Q** as at **T**.

14 9702/01/M/J/05/Q4*

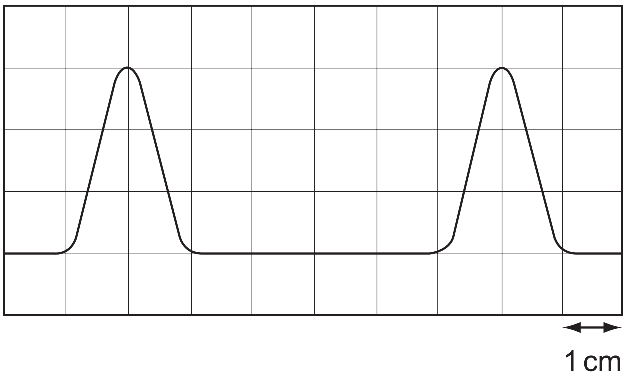
In an experiment, a radio-controlled car takes 2.50 ± 0.05 s to travel 40.0 ± 0.1 m.

What is the car's average speed and the uncertainty in this value?

A $16 \pm 1\,\mathrm{m\,s^{-1}}$ **B** $16.0 \pm 0.2\,\mathrm{m\,s^{-1}}$ **C** $16.0 \pm 0.4\,\mathrm{m\,s^{-1}}$ **D** $16.00 \pm 0.36\,\mathrm{m\,s^{-1}}$

15 9702/01/M/J/05/Q5

The diagram shows two pulses on the screen of a cathode ray oscilloscope. A grid of 1 cm squares covers the screen. The time base setting is $1\,\mu\mathrm{s\,cm^{-1}}$.



How long does each pulse last?

A 2 μs **B** 3 μs

C 4 μs **D** 6 μs

16 9702/01/M/J/05/Q6

Which feature of a graph allows acceleration to be determined?

A the area under a displacement-time graph

B the area under a velocity-time graph

C the slope of a displacement-time graph

D the slope of a velocity-time graph

17 9702/01/M/J/05/Q7

A boy throws a ball vertically upwards. It rises to a maximum height, where it is momentarily at rest, and falls back to his hands.

Which of the following gives the acceleration of the ball at various stages in its motion? Take vertically upwards as positive. Neglect air resistance.

	rising	at maximum height	falling
A	$-9.81\,\mathrm{m\,s^{-2}}$	0	$+9.81\,\mathrm{m\,s^{-2}}$
B	$-9.81\,\mathrm{m\,s^{-2}}$	$-9.81\,\mathrm{m\,s^{-2}}$	$-9.81\,\mathrm{m\,s^{-2}}$
C	$+9.81\,\mathrm{m\,s^{-2}}$	$+9.81\,\mathrm{m\,s^{-2}}$	$+9.81\,\mathrm{m\,s^{-2}}$
D	$+9.81\,\mathrm{m\,s^{-2}}$	0	$-9.81\,\mathrm{m\,s^{-2}}$

18 **9702/01/O/N/05/Q4**

A steel rule can be read to the nearest millimetre. It is used to measure the length of a bar whose true length is 895 mm. Repeated measurements give the following readings.

length / mm 892, 891, 892, 891, 891, 892

Are the readings accurate and precise to within 1 mm?

	results are accurate to within 1 mm	results are precise to within 1 mm
A	no	no
B	no	yes
C	yes	no
D	yes	yes

19 **9702/01/O/N/05/Q5**

The density of the material of a rectangular block is determined by measuring the mass and linear dimensions of the block. The table shows the results obtained, together with their uncertainties.

mass	=	(25.0 ± 0.1) g	breadth	=	(2.00 ± 0.01) cm
length	=	(5.00 ± 0.01) cm	height	=	(1.00 ± 0.01) cm

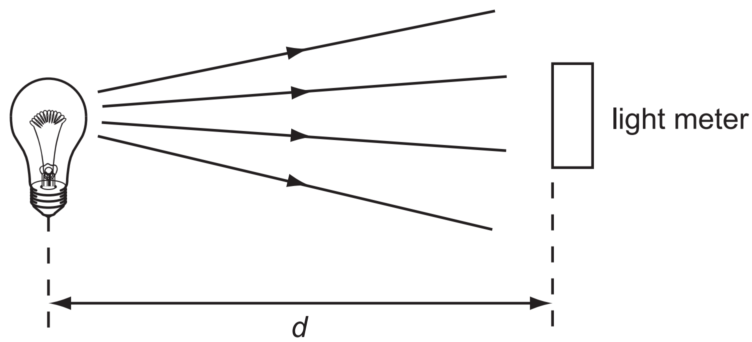
The density is calculated to be 2.50 g cm^{-3}.

What is the uncertainty in this result?

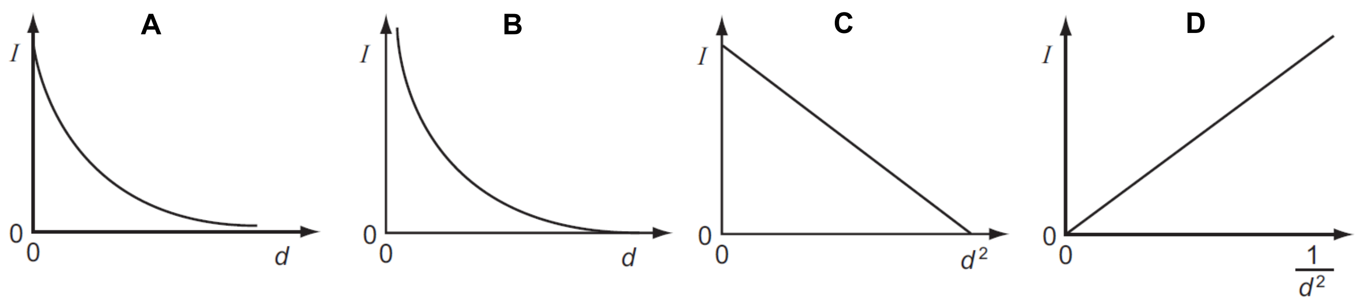
A ± 0.01 g cm^{-3} **B** ± 0.02 g cm^{-3} **C** ± 0.05 g cm^{-3} **D** ± 0.13 g cm^{-3}

20 **9702/01/M/J/06/Q4**

A light meter measures the intensity I of the light falling on it. Theory suggests that this varies as the inverse square of the distance d.

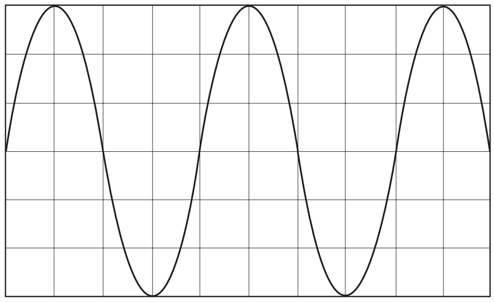


Which graph of the results supports this theory?



21 9702/01/M/J/06/Q5

The cathode-ray oscilloscope (c.r.o.) display shows the waveform produced by an electronic circuit. The c.r.o. time-base is set at 10 ms per division.



What is the period of the signal shown?

A 20 ms **B** 30 ms

C 40 ms **D** 80 ms

22 9702/01/M/J/06/Q6

The resistance R of an unknown resistor is found by measuring the potential difference V across the resistor and the current I through it and using the equation $R = \frac{V}{I}$. The voltmeter reading has a 3 % uncertainty and the ammeter reading has a 2 % uncertainty.

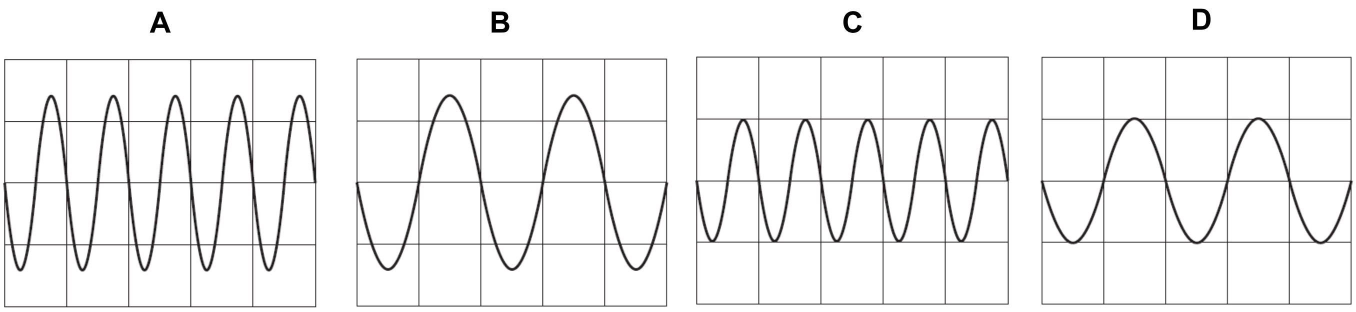
What is the uncertainty in the calculated resistance?

A 1.5 % **B** 3 % **C** 5 % **D** 6 %

23 9702/01/O/N/06/Q4*

The Y-input terminals of a cathode-ray oscilloscope (c.r.o.) are connected to a supply of peak value 5.0 V and of frequency 50 Hz. The time-base is set at 10 ms per division and the Y-gain at 5.0 V per division.

Which trace is obtained?



24 9702/01/O/N/06/Q5

The measurement of a physical quantity may be subject to random errors and to systematic errors.

Which statement is correct?

A Random errors can be reduced by taking the average of several measurements.

B Random errors are always caused by the person taking the measurement.

C A systematic error cannot be reduced.

D A systematic error results in a different reading each time the measurement is taken.

25 9702/01/O/N/06/Q6

An experiment is done to measure the resistance of a wire.

The current in the wire is 1.0 ± 0.2 A and the potential difference across the wire is 8.0 ± 0.4 V.

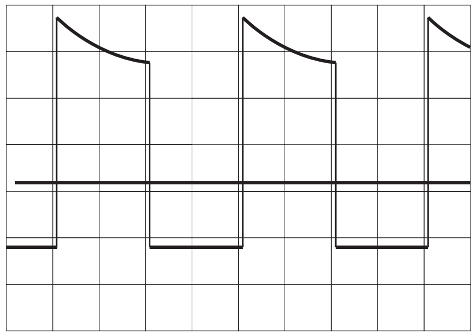
What is the resistance of the wire and its uncertainty?

A $(8.0 \pm 0.2)\,\Omega$ **B** $(8.0 \pm 0.6)\,\Omega$ **C** $(8 \pm 1)\,\Omega$ **D** $(8 \pm 2)\,\Omega$

26 **9702/01/M/J/07/Q4**

An oscilloscope display consists of two separate traces, a waveform and a long horizontal line. The horizontal line may be taken as the zero level.

The grid on the screen is calibrated in cm squares, the timebase setting is 2.5 ms cm^{-1}, and the Y-sensitivity is 5 mV cm^{-1}.

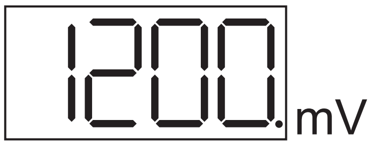


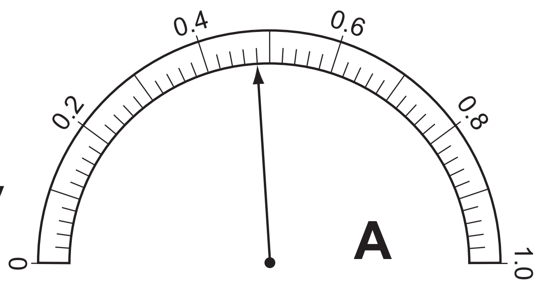
What are the period and the peak positive voltage of the waveform in the diagram?

	period / ms	peak positive voltage / mV
A	5	17
B	5	25
C	10	17
D	10	25

27 **9702/01/M/J/07/Q5**

The resistance of an electrical component is measured. The following meter readings are obtained.





What is the resistance?

A 2.5 Ω **B** 2.7 Ω

C 2500 Ω **D** 2700 Ω

28 **9702/01/O/N/07/Q4**

A series of measurements of the acceleration of free fall *g* is shown in the table.

Which set of results is precise but not accurate?

	g / m s^{-2}				
A	9.81	9.79	9.84	9.83	9.79
B	9.81	10.12	9.89	8.94	9.42
C	9.45	9.21	8.99	8.76	8.51
D	8.45	8.46	8.50	8.41	8.47

29 **9702/01/O/N/07/Q5**

A mass *m* has acceleration *a*. It moves through a distance *s* in time *t*. The power used in accelerating the mass is equal to the product of force and velocity. The percentage uncertainties are

0.1 % in *m*,

1 % in *a*,

1.5 % in *s*,

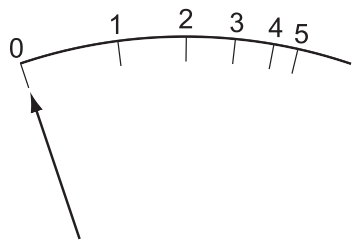
0.5 % in *t*.

What is the percentage uncertainty in the average power?

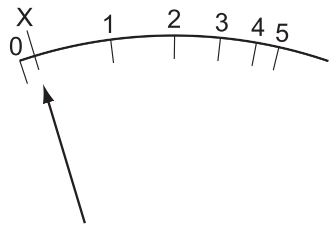
A 2.1 % **B** 2.6 % **C** 3.1 % **D** 4.1 %

30 9702/01/O/N/07/Q6

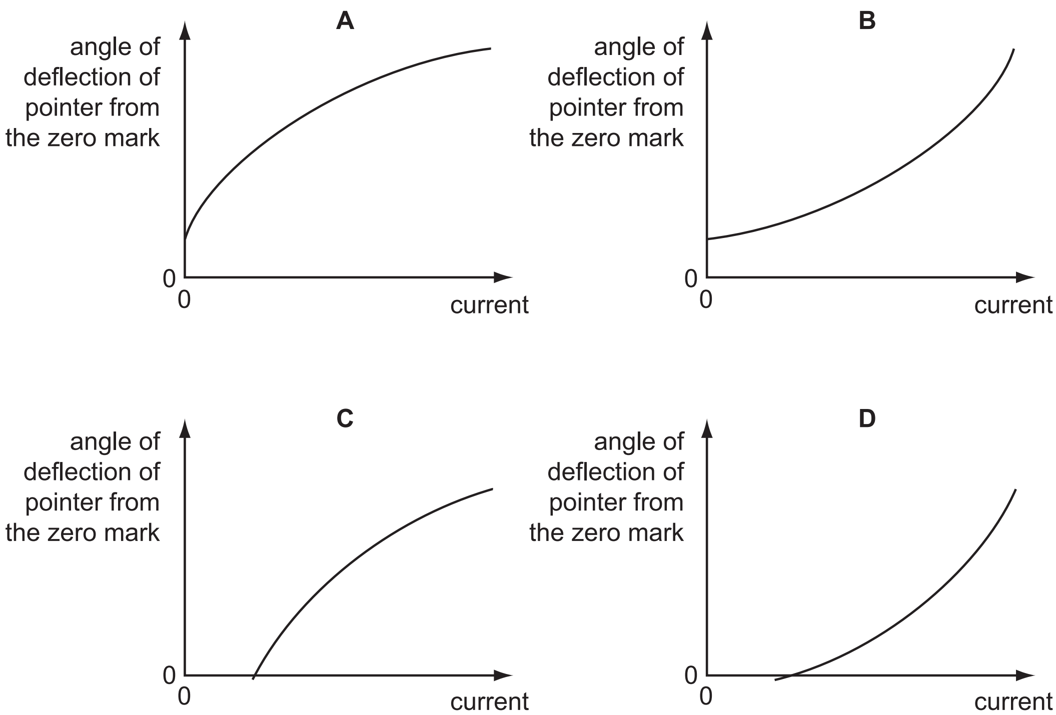
The diagram shows the graduations of a correctly calibrated ammeter. When the current is zero, the pointer is at 0.



The ammeter is accidentally readjusted so that when the current is zero, the pointer is at X.



Which calibration graph best represents the response of the readjusted ammeter?



31 9702/01/M/J/08/Q4

The resistance R of a resistor is determined by measuring the potential difference V across it and the current I in it. The value of R is then calculated using the equation

$$R = \frac{V}{I}.$$

The values measured are $V = 1.00 \pm 0.05\,\text{V}$ and $I = 0.50 \pm 0.01\,\text{A}$.

What is the percentage uncertainty in the value of R?

A 2.5% **B** 3.0% **C** 7.0% **D** 10.0%

32 9702/01/M/J/08/Q5

Four students each made a series of measurements of the acceleration of free fall g. The table shows the results obtained.

Which set of results could be described as precise but **not** accurate?

	$g / \mathrm{m\,s^{-2}}$			
A	9.81	9.79	9.84	9.83
B	9.81	10.12	9.89	8.94
C	9.45	9.21	8.99	8.76
D	8.45	8.46	8.50	8.41

33 9702/01/O/N/08/Q4

A student uses a digital ammeter to measure a current. The reading of the ammeter is found to fluctuate between 1.98 A and 2.02 A.

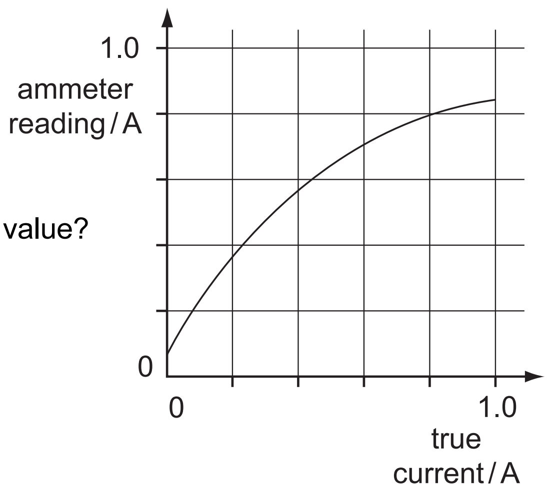
The manufacturer of the ammeter states that any reading has a systematic uncertainty of ±1%.

Which value of current should be quoted by the student?

A (2.00 ± 0.01) A **B** (2.00 ± 0.02) A **C** (2.00 ± 0.03) A **D** (2.00 ± 0.04) A

34 9702/11/M/J/16/Q5

A calibration graph is produced for a faulty ammeter.



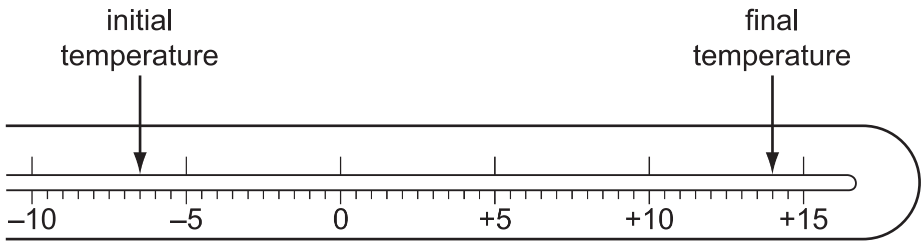
Which ammeter reading will be nearest to the correct value?

A 0.2 A **B** 0.4 A

C 0.6 A **D** 0.8 A

35 9702/13/M/J/13/Q6

The diagram shows the stem of a Celsius thermometer marked to show initial and final temperature values.

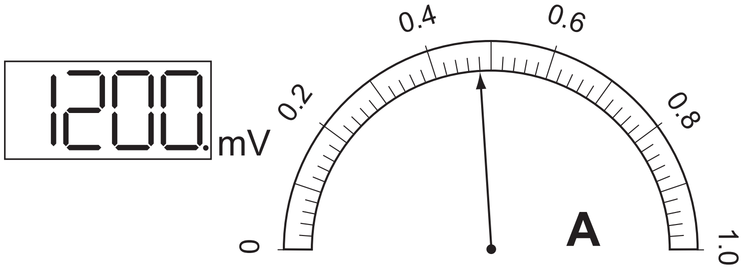


What is the temperature change expressed to an appropriate number of significant figures?

A 14 °C **B** 20.5 °C **C** 21 °C **D** 22.0 °C

36 **9702/01/M/J/09/Q4**

The diagrams show digital voltmeter and analogue ammeter readings from a circuit in which electrical heating is occurring.



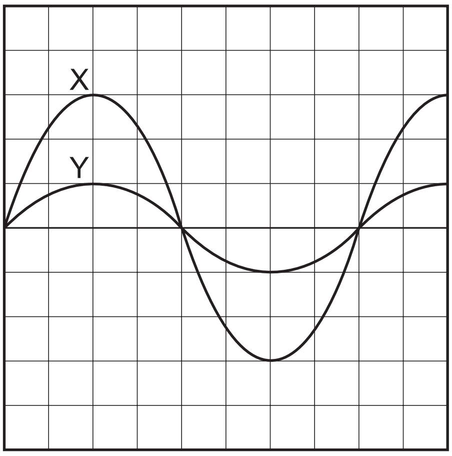
What is the electrical power of the heater?

A 0.53 W **B** 0.58 W

C 530 W **D** 580 W

37 **9702/12/O/N/09/Q3**

The diagram shows an oscilloscope screen displaying two signals.



Signal X has a frequency of 50 Hz and peak voltage of 12 V.

What is the period and peak voltage of signal Y?

	period / ms	peak voltage / V
A	20	4
B	20	12
C	50	4
D	50	12

38 **9702/12/M/J/10/Q3**

A student finds the density of a liquid by measuring its mass and its volume. The following is a summary of his measurements.

mass of empty beaker = (20 ± 1) g

mass of beaker + liquid = (70 ± 1) g

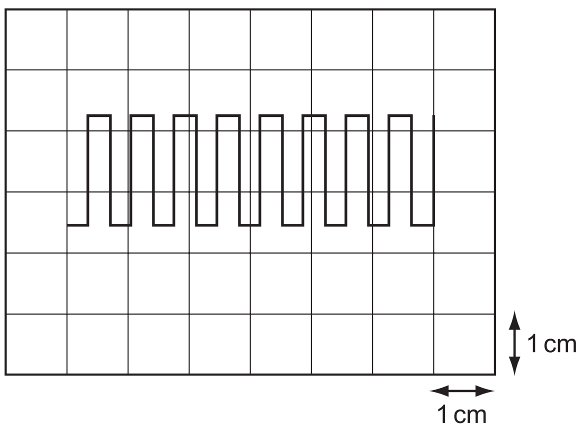
volume of liquid = $(10.0 \pm 0.6)\,\text{cm}^3$

He correctly calculates the density of the liquid as $5.0\,\text{g cm}^{-3}$.
What is the uncertainty in this value?

A $0.3\,\text{g cm}^{-3}$ **B** $0.5\,\text{g cm}^{-3}$ **C** $0.6\,\text{g cm}^{-3}$ **D** $2.6\,\text{g cm}^{-3}$

39 **9702/12/M/J/10/Q5**

The diagram shows a square-wave trace on the screen of a cathode-ray oscilloscope. A grid of 1 cm squares covers the screen. The time-base setting is $10\,\text{ms cm}^{-1}$.



What is the approximate frequency of the square wave?

A 70 Hz **B** 140 Hz

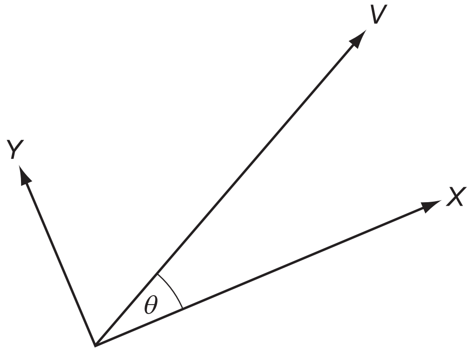
C 280 Hz **D** 1400 Hz

40 **9702/12/M/J/10/Q6**

A vector quantity V is resolved into two perpendicular components X and Y. The angle between V and component X is θ.

The angle between component X and the vector V is increased from 0° to 90°.

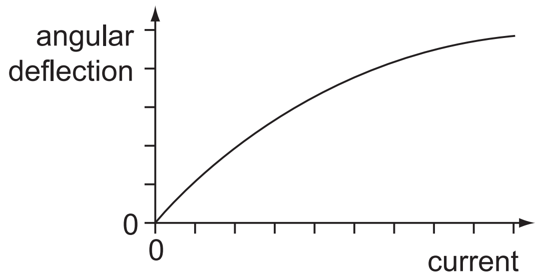
How do the magnitudes of X and Y change as the angle θ is increased in this way?



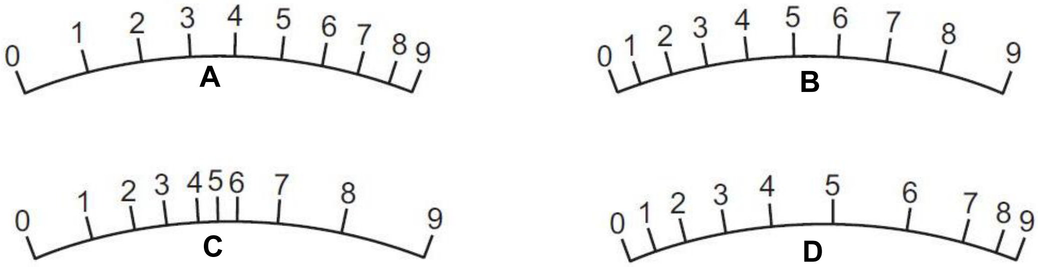
	X	Y
A	increase	increase
B	increase	decrease
C	decrease	increase
D	decrease	decrease

41 **9702/11/O/N/10/Q4**

The angular deflection of the needle of an ammeter varies with the current passing through the ammeter as shown in the graph.

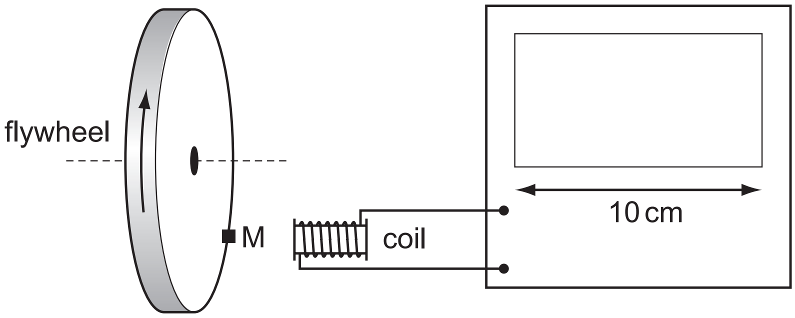


Which diagram could represent the appearance of the scale on this meter?



42 **9702/11/O/N/10/Q5**

The diagram shows a cathode-ray oscilloscope (c.r.o.) being used to measure the rate of rotation of a flywheel.



The flywheel has a small magnet M mounted on it. Each time the magnet passes the coil, a voltage pulse is generated, which is passed to the c.r.o. The display of the c.r.o. is 10 cm wide. The flywheel is rotating at a rate of about 3000 revolutions per minute.

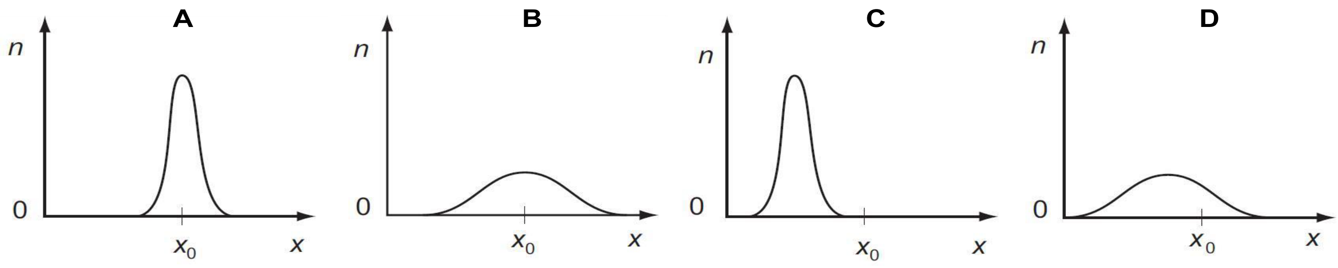
Which time-base setting will display clearly separate pulses on the screen?

A $1\,\text{s}\,\text{cm}^{-1}$ **B** $10\,\text{ms}\,\text{cm}^{-1}$ **C** $100\,\mu\text{s}\,\text{cm}^{-1}$ **D** $1\,\mu\text{s}\,\text{cm}^{-1}$

43 **9702/11/O/N/10/Q6**

A fixed quantity x_0 is measured many times in an experiment that has experimental uncertainty. A graph is plotted to show the number n of times that a particular value x is obtained.

Which graph could be obtained if the measurement of x_0 has a large systematic error but a small random error?



44 **9702/12/O/N/10/Q4**

A metre rule is used to measure the length of a piece of wire. It is found to be 70 cm long to the nearest millimetre.

How should this result be recorded in a table of results?

A 0.7 m **B** 0.70 m **C** 0.700 m **D** 0.7000 m

45 **9702/12/O/N/10/Q5**

A quantity x is to be determined from the equation

$$x = P - Q.$$

P is measured as 1.27 ± 0.02 m.

Q is measured as 0.83 ± 0.01 m.

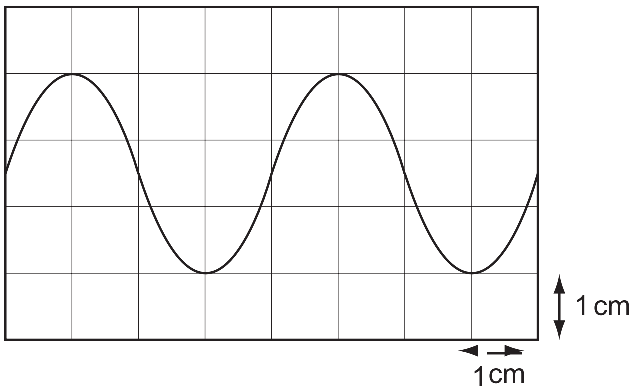
What is the percentage uncertainty in x to one significant figure?

A 0.4 % **B** 2 % **C** 3 % **D** 7 %

46 **9702/11/M/J/11/Q4**

The diagram shows a trace of a wave on a cathode-ray oscilloscope.

The vertical and horizontal gridlines have a spacing of 1.0 cm. The voltage scaling is $4\,\mathrm{V\,cm^{-1}}$ and the time scaling is $5\,\mathrm{ms\,cm^{-1}}$.

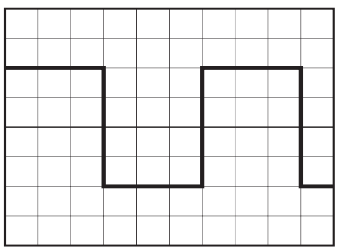


What are the amplitude and period of the wave?

	amplitude / V	period / ms
A	1.5	4
B	5.0	10
C	6.0	20
D	12.0	20

47 **9702/13/O/N/12/Q5**

A cathode-ray oscilloscope displays a square wave, as shown in the diagram.



The time-base setting is 0.20 ms per division.

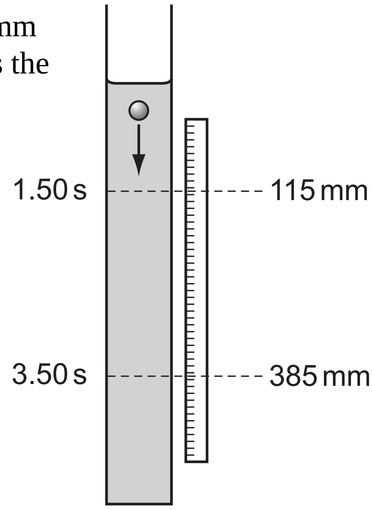
What is the frequency of the square wave?

A 8.3 Hz **B** 830 Hz

C 1300 Hz **D** 1700 Hz

48 **9702/11/M/J/11/Q5**

The diagram shows an experiment to measure the speed of a small ball falling at constant speed through a clear liquid in a glass tube.



There are two marks on the tube. The top mark is positioned at 115 ± 1 mm on the adjacent rule and the lower mark at 385 ± 1 mm. The ball passes the top mark at 1.50 ± 0.02 s and passes the lower mark at 3.50 ± 0.02 s.

The constant speed of the ball is calculated by

$$\frac{385-115}{3.50-1.50} = \frac{270}{2.00} = 135\,\text{mm}\,\text{s}^{-1}.$$

Which expression calculates the fractional uncertainty in the value of this speed?

A $\dfrac{2}{270} + \dfrac{0.04}{2.00}$ **B** $\dfrac{2}{270} - \dfrac{0.04}{2.00}$

C $\dfrac{1}{270} \times \dfrac{0.02}{2.00}$ **D** $\dfrac{1}{270} \div \dfrac{0.02}{2.00}$

49 **9702/12/M/J/11/Q4**

The uncertainty in the value of the momentum of a trolley passing between two points X and Y varies with the choice of measuring devices.

Measurements for the same trolley made by different instruments were recorded.

1. distance between X and Y using a metre rule with cm divisions = 0.55 m
2. distance between X and Y using a metre rule with mm divisions = 0.547 m
3. timings using a wristwatch measuring to the nearest 0.5 s at X = 0.0 s and at Y = 4.5 s
4. timings using light gates measuring to the nearest 0.1 s at X = 0.0 s and at Y = 4.3 s
5. mass of trolley using a balance measuring to the nearest g = 6.4×10^{-2} kg
6. mass of trolley using a balance measuring to the nearest 10 g = 6×10^{-2} kg

Which measurements, one for each quantity measured, lead to the least uncertainty in the value of the momentum of the trolley?

A 1, 3 and 6 **B** 1, 4 and 6 **C** 2, 3 and 6 **D** 2, 4 and 5

50 **9702/12/M/J/11/Q6**

A bullet is fired horizontally with speed v from a rifle. For a short time t after leaving the rifle, the only force affecting its motion is gravity. The acceleration of free fall is g.

Which expression gives the value of $\dfrac{\text{the horizontal distance travelled in time } t}{\text{the vertical distance travelled in time } t}$?

A $\dfrac{vt}{g}$ **B** $\dfrac{v}{gt}$ **C** $\dfrac{2vt}{g}$ **D** $\dfrac{2v}{gt}$

51 **9702/12/O/N/11/Q4**

A micrometer is used to measure the diameters of two cylinders.

diameter of first cylinder = 12.78 ± 0.02 mm

diameter of second cylinder = 16.24 ± 0.03 mm

The difference in the diameters is calculated.

What is the uncertainty in this difference?

A ± 0.01 mm **B** ± 0.02 mm **C** ± 0.03 mm **D** ± 0.05 mm

52 9702/13/O/N/15/Q6

A cylindrical tube rolling down a slope of inclination θ moves a distance L in time T. The equation relating these quantities is

$$L\left(3+\frac{a^2}{P}\right) = QT^2 \sin\theta$$

Where a is the internal radius of the tube and P and Q are constants.

Which line gives the correct units for P and Q?

	P	Q
A	m^2	$m^2 s^{-2}$
B	m^2	$m s^{-2}$
C	m^2	$m^3 s^{-2}$
D	m^3	$m s^{-2}$

53 9702/12/O/N/11/Q5

The speedometer in a car consists of a pointer which rotates. The pointer is situated several millimetres from a calibrated scale.

What could cause a random error in the driver's measurement of the car's speed?

A The car's speed is affected by the wind direction.

B The driver's eye is not always in the same position in relation to the pointer.

C The speedometer does not read zero when the car is at rest.

D The speedometer reads 10 % higher than the car's actual speed.

54 9702/12/M/J/12/Q2

For which quantity is the magnitude a reasonable estimate?

A frequency of a radio wave 500 pHz

B mass of an atom 500 μg

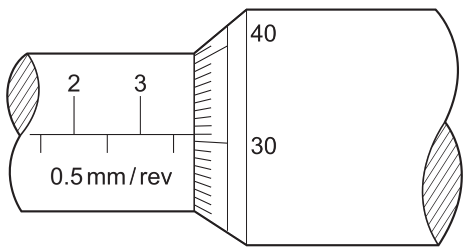
C the Young modulus of a metal 500 kPa

D wavelength of green light 500 nm

55 9702/12/M/J/12/Q4*

The diameter of a cylindrical metal rod is measured using a micrometer screw gauge.

The diagram below shows an enlargement of the scale on the micrometer screw gauge when taking the measurement.



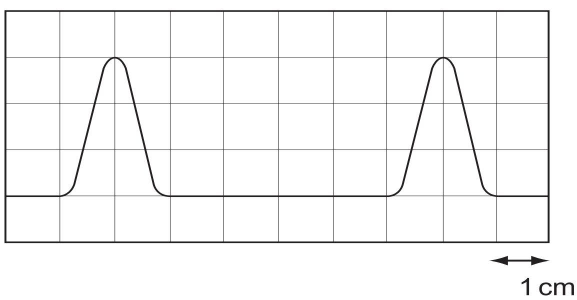
What is the cross-sectional area of the rod?

A $3.81\,mm^2$ **B** $11.4\,mm^2$

C $22.8\,mm^2$ **D** $45.6\,mm^2$

56 9702/12/M/J/12/Q6

The diagram shows two complete pulses on the screen of a cathode-ray oscilloscope. A grid of 1 cm squares covers the screen. The time-base setting is $1\,\mu s\,cm^{-1}$.



How long does each pulse last?

A 2 μs **B** 3 μs

C 4 μs **D** 6 μs

57 **9702/12/M/J/12/Q5**

A mass is dropped from rest, and falls through a distance of 2.0 m in a vacuum. An observer records the time taken for the mass to fall through this distance using a manually operated stopwatch and repeats the measurements a further two times. The average result of these measured times, displayed in the table below, was used to determine a value for the acceleration of free fall. This was calculated to be $9.8\,\text{m s}^{-2}$.

	first measurement	second measurement	third measurement	average
time / s	0.6	0.73	0.59	0.64

Which statement best relates to the experiment?

A The measurements are precise and accurate with no evidence of random errors.

B The measurements are not accurate and not always recorded to the degree of precision of the measuring device but the calculated experimental result is accurate.

C The measurements are not always recorded to the degree of precision of the measuring device but are accurate. Systematic errors may be present.

D The range of results shows that there were random errors made but the calculated value is correct so the experiment was successful.

58 **9702/13/M/J/12/Q4***

In an experiment, a radio-controlled car takes 2.50 ± 0.05 s to travel 40.0 ± 0.1 m.

What is the car's average speed and the uncertainty in this value?

A $16 \pm 1\,\text{m s}^{-1}$ **B** $16.0 \pm 0.2\,\text{m s}^{-1}$ **C** $16.0 \pm 0.4\,\text{m s}^{-1}$ **D** $16.00 \pm 0.36\,\text{m s}^{-1}$

A student is given a reel of wire of diameter less than 0.2 mm and is asked to find the density of the metal.

Which pair of instruments would be most suitable for finding the **volume** of the wire?

A balance and micrometer **C** metre rule and vernier calipers

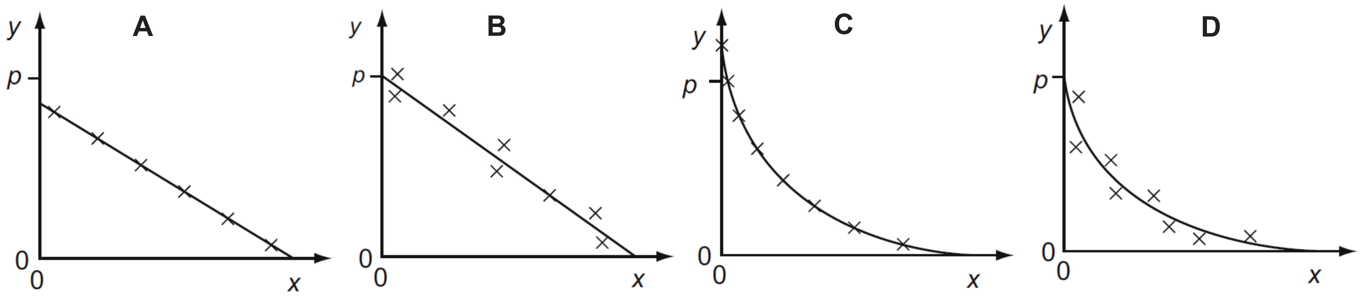
B metre rule and micrometer **D** micrometer and vernier calipers

59 **9702/12/O/N/12/Q6**

Variables x and y are related by the equation $y = p - qx$ where p and q are constants.

Values of x and y are measured experimentally. The results contain a systematic error.

Which graph best represents these results?



60 **9702/12/O/N/12/Q8**

A science museum designs an experiment to show the fall of a feather in a vertical glass vacuum tube.

The time of fall from rest is to be close to 0.5 s.

What length of tube is required?

A 1.3 m **B** 2.5 m **C** 5.0 m **D** 10.0 m

61 **9702/12/O/N/12/Q7**

The speed of a car is calculated from measurements of the distance travelled and the time taken.

The distance is measured as 200 m, with an uncertainty of ± 2 m.

The time is measured as 10.0 s, with an uncertainty of ± 0.2 s.

What is the percentage uncertainty in the calculated speed?

A ± 0.5 % **B** ± 1 % **C** ± 2 % **D** ± 3 %

62 **9702/13/O/N/12/Q6***

What will reduce the systematic errors when taking a measurement?

A adjusting the needle on a voltmeter so that it reads zero when there is no potential difference across it

B measuring the diameter of a wire at different points and taking the average

C reducing the parallax effects by using a marker and a mirror when measuring the amplitude of oscillation of a pendulum

D timing 20 oscillations, rather than a single oscillation, when finding the period of a pendulum

63 **9702/13/O/N/12/Q7**

In an experiment to determine the acceleration of free fall g, the time t taken for a ball to fall through distance s was measured. The uncertainty in the measurement of s is estimated to be 2 %. The uncertainty in the measurement of t is estimated to be 3 %.

The value of g is determined using the equation $g = \dfrac{2s}{t^2}$.

What is the uncertainty in the calculated value of g?

A 1 % **B** 5 % **C** 8 % **D** 11 %

64 **9702/11/M/J/13/Q5**

In an experiment to determine the acceleration of free fall g, the period of oscillation T and length l of a simple pendulum were measured. The uncertainty in the measurement of l is estimated to be 4%, and the uncertainty in the measurement of T is estimated to be 1%.

The value of g is determined using the formula $g = \dfrac{4\pi^2 l}{T^2}$.

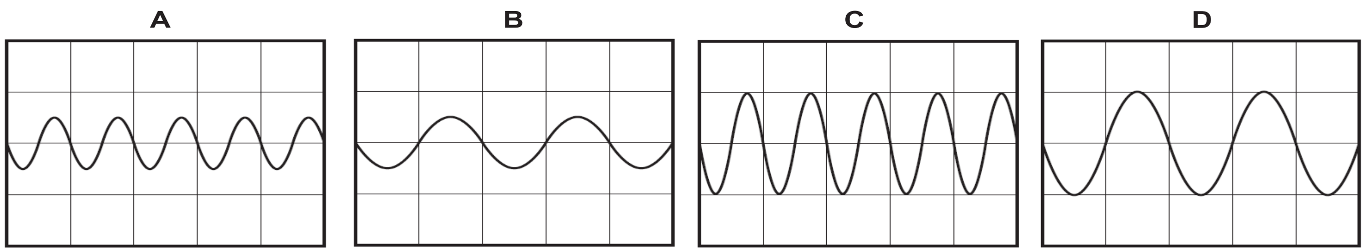
What is the uncertainty in the calculated value for g?

A 2% **B** 3% **C** 5% **D** 6%

65 **9702/11/M/J/13/Q6**

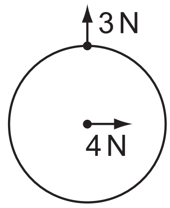
The Y-input terminals of a cathode-ray oscilloscope (c.r.o.) are connected to a supply of amplitude 5.0 V and frequency 50 Hz. The time-base is set at 10 ms per division and the Y-gain at 5.0 V per division.

Which trace is obtained?

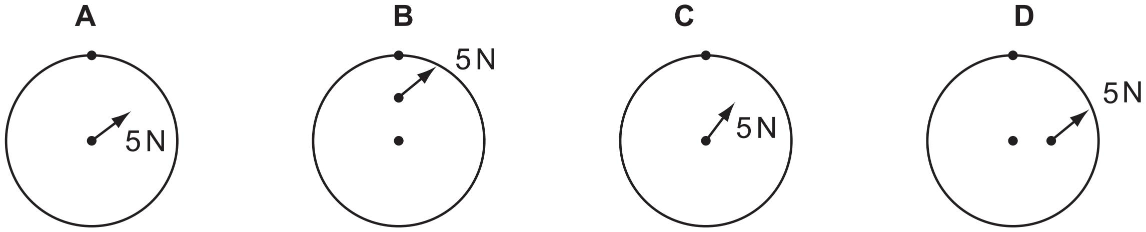


66 9702/12/M/J/13/Q3

Two forces act on a circular disc as shown.



Which diagram shows the line of action of the resultant force?



67 9702/12/M/J/13/Q4

A student carried out an experiment in which an electric current was known to decrease with time. The readings he found, from first to last, were 3.62 mA, 2.81 mA, 1.13 mA, 1.76 mA and 0.90 mA.

Which statement could **not** explain the anomalous 1.13 mA reading?

A He has reversed the third and fourth readings in the results table.

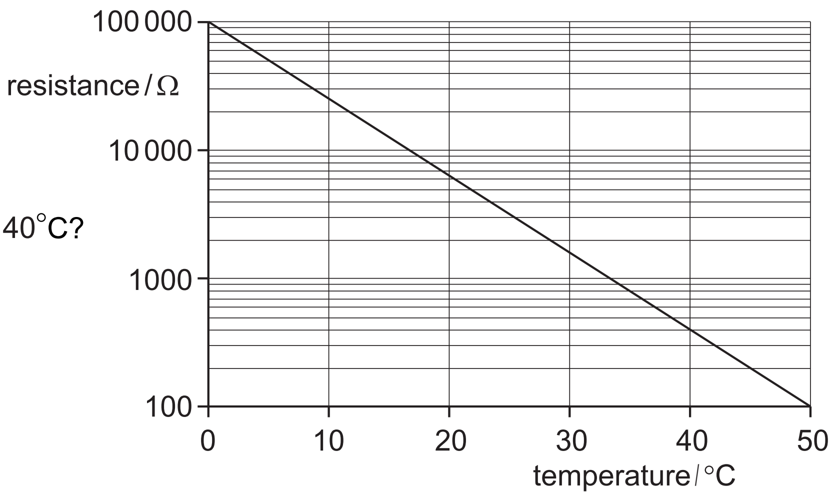
B He read the ammeter incorrectly; the reading should have been 2.13 mA.

C He took the current reading at the wrong time.

D There was a systematic error in the readings from the ammeter.

68 9702/12/M/J/13/Q5

The diagram shows a calibration curve for a thermistor, drawn with an unusual scale on the vertical axis.



What is the thermistor resisitance corresponding to a temperature of 40°C?

A 130 Ω **B** 150 Ω

C 400 Ω **D** 940 Ω

69 9702/13/M/J/13/Q3

Which statement is **incorrect** by a factor of 100 or more?

A Atmospheric pressure is about 1×10^5 Pa.

B Light takes 5×10^2 s to reach us from the Sun.

C The frequency of ultra-violet light is 3×10^{12} Hz.

D The life-span of a man is about 2×10^9 s.

70 9702/13/M/J/13/Q5

A student takes measurements of the current in a resistor of constant resistance and the potential difference (p.d.) across it. The readings are then used to plot a graph of current against p.d.

There is a systematic error in the current readings.

How could this be identified from the graph?

A At least one anomalous data point can be identified.

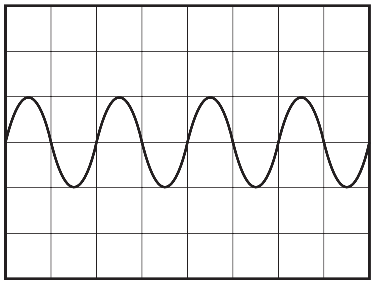
B The data points are scattered about the straight line of best fit.

C The graph is a curve, not a straight line.

D The straight line graph does not pass through the origin.

71 9702/11/M/J/15/Q4

A whale produces sound waves of frequency 5 Hz. The waves are detected by a microphone and displayed on an oscilloscope.



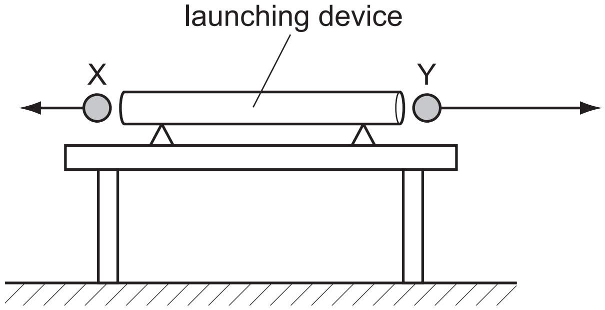
What is the time-base setting on the oscilloscope?

A $0.1\,\text{ms div}^{-1}$ **B** $1\,\text{ms div}^{-1}$

C $10\,\text{ms div}^{-1}$ **D** $100\,\text{ms div}^{-1}$

72 9702/13/M/J/13/Q7

A double-ended launching device fires two identical steel balls X and Y at exactly the same time. The diagram shows the initial velocities of the balls. They are both launched horizontally, but Y has greater speed.



Which statement explains what an observer would see?

A Both X and Y reach the ground simultaneously, because air resistance will cause both to have the same final speed.

B Both X and Y reach the ground simultaneously, because gravitational acceleration is the same for both.

C X reaches the ground before Y, because X lands nearer to the launcher.

D Y reaches the ground before X, because Y has greater initial speed.

73 9702/11/M/J/14/Q4

An experiment is carried out to measure the resistance of a wire.

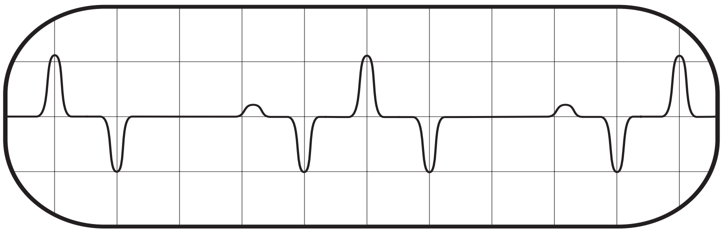
The current in the wire is (1.0 ± 0.2) A and the potential difference across the wire is (8.0 ± 0.4) V.

What is the resistance of the wire and its uncertainty?

A $(8.0 \pm 0.2)\,\Omega$ **B** $(8.0 \pm 0.6)\,\Omega$ **C** $(8 \pm 1)\,\Omega$ **D** $(8 \pm 2)\,\Omega$

74 **9702/12/O/N/13/Q4**

A signal that repeats periodically is displayed on the screen of a cathode-ray oscilloscope.



The screen has 1 cm squares and the time base is set at 2.00 ms cm^{-1}.

What is the frequency of this periodic signal?

A 50 Hz **B** 100 Hz

C 125 Hz **D** 200 Hz

75 **9702/12/O/N/13/Q5**

A micrometer screw gauge is used to measure the diameter of a small uniform steel sphere. The micrometer reading is 5.00 mm ± 0.01 mm.

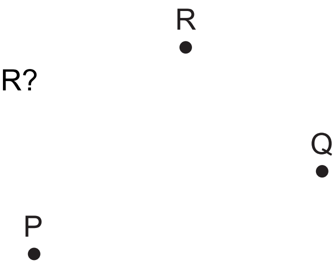
What will be the percentage uncertainty in a calculation of the volume of the sphere, using these values?

A 0.2% **B** 0.4% **C** 0.6% **D** 1.2%

76 **9702/12/O/N/13/Q6**

One object moves directly from P to R.

In a shorter time, a second object moves from P to Q to R.



Which statement about the two objects is correct for the journey from P to R?

A They have the same average speed.

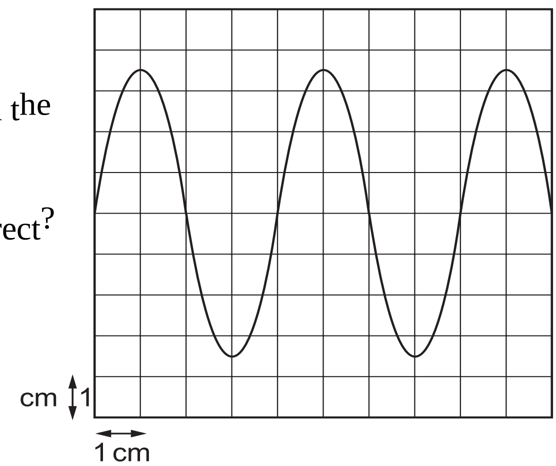
B They have the same average velocity.

C They have the same displacement.

D They travel the same distance.

77 **9702/12/M/J/14/Q3***

A cathode-ray oscilloscope (c.r.o.) is connected to an alternating voltage. The following trace is produced on the screen.



The oscilloscope time-base setting is 0.5 ms cm^{-1} and the Y-plate sensitivity is 2 V cm^{-1}.

Which statement about the alternating voltage is correct?

A The amplitude is 3.5 cm.

B The frequency is 0.5 kHz.

C The period is 1 ms.

D The wavelength is 4 cm.

78 **9702/13/O/N/13/Q6**

A student wishes to determine the density ρ of lead. She measures the mass and diameter of a small sphere of lead:

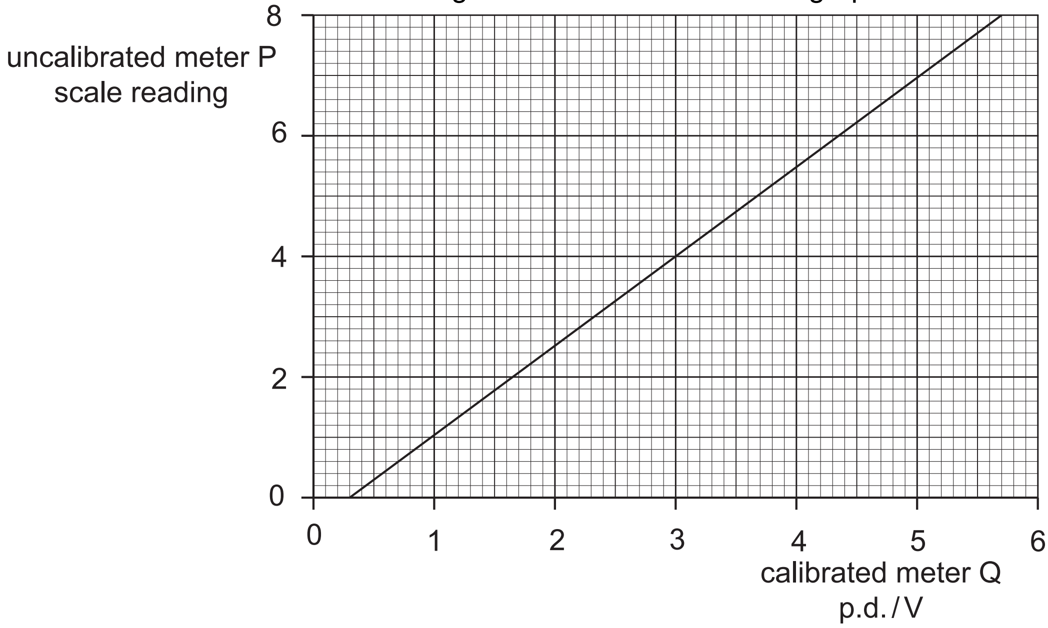
mass = (0.506 ± 0.005) g diameter = (2.20 ± 0.02) mm.

What is the best estimate of the percentage uncertainty in her value of ρ?

A 1.9% **B** 2.0% **C** 2.8% **D** 3.7%

79 9702/13/O/N/13/Q5

An uncalibrated analogue voltmeter P is connected in parallel with another voltmeter Q which is known to be accurately calibrated. For a range of values of potential difference (p.d.), readings are taken from the two meters. The diagram shows the calibration graph obtained.



The graph shows that meter P has a zero error. This meter is now adjusted to remove this zero error. When the meter is recalibrated, the gradient of the calibration graph is found to be unchanged.

What is the new scale reading on meter P, when it is used to measure a p.d. of 5.0 V?

The graph shows that meter P has a zero error. This meter is now adjusted to remove this zero error. When the meter is recalibrated, the gradient of the calibration graph is found to be unchanged.

What is the new scale reading on meter P when it is used to measure a p.d. of 5.0 V?

A 6.6 **B** 6.7 **C** 7.2 **D** 7.4

80 9702/11/M/J/14/Q5

The Young modulus of the material of a wire is to be found. The Young modulus E is given by the equation below.

$$E = \frac{4Fl}{\pi d^2 x}$$

The wire is extended by a known force and the following measurements are made.

Which measurement has the largest effect on the uncertainty in the value of the calculated Young modulus?

	measurement	symbol	value
A	length of wire before force applied	l	2.043 ± 0.002 m
B	diameter of wire	d	0.54 ± 0.02 mm
C	force applied	F	19.62 ± 0.01 N
D	extension of wire with force applied	x	5.2 ± 0.2 mm

81 9702/12/M/J/14/Q4

A quantity y is to be determined from the equation shown. $y = \dfrac{px}{q^2}$

The percentage uncertainties in p, x and q are shown.

	percentage uncertainty
p	6%
x	2%
q	4%

What is the percentage uncertainty in y?

A 0.5 % **B** 1 %

C 16 % **D** 192 %

82 9702/12/M/J/14/Q5

A thermometer can be read to an accuracy of ±0.5 °C. This thermometer is used to measure a temperature rise from 40 °C to 100 °C.

What is the percentage uncertainty in the measurement of the temperature rise?

A 0.5 % **B** 0.8 % **C** 1.3 % **D** 1.7 %

83 9702/13/M/J/14/Q4*

The resistance of a lamp is calculated from the value of the potential difference (p.d.) across it and the value of the current passing through it.

Which statement correctly describes how to combine the uncertainties in the p.d. and in the current?

A Add together the actual uncertainty in the p.d. and the actual uncertainty in the current.

B Add together the percentage uncertainty in the p.d. and the percentage uncertainty in the current.

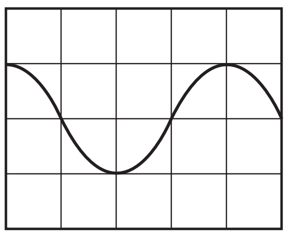
C Subtract the actual uncertainty in the current from the actual uncertainty in the p.d.

D Subtract the percentage uncertainty in the current from the percentage uncertainty in the p.d.

84 9702/13/M/J/14/Q5*

The display on a cathode-ray oscilloscope shows the signal produced by an electronic circuit. The time-base is set at 5.0 ns per division and the Y-gain at 10 V per division.

What is the frequency of the signal?



A 2.0×10^{-8} Hz **B** 2.5×10^{-2} Hz

C 5.0×10^{7} Hz **D** 3.1×10^{8} Hz

85 9702/13/M/J/14/Q4

The resistance of a lamp is calculated from the value of the potential difference (p.d.) across it and the value of the current passing through it.

Which statement correctly describes how to combine the uncertainties in the p.d. and in the current?

A Add together the actual uncertainty in the p.d. and the actual uncertainty in the current.

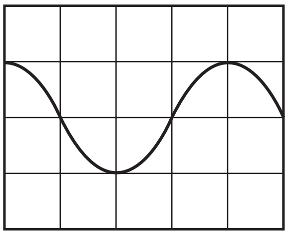
B Add together the percentage uncertainty in the p.d. and the percentage uncertainty in the current.

C Subtract the actual uncertainty in the current from the actual uncertainty in the p.d.

D Subtract the percentage uncertainty in the current from the percentage uncertainty in the p.d.

86 **9702/13/M/J/14/Q5**

The display on a cathode-ray oscilloscope shows the signal produced by an electronic circuit. The time-base is set at 5.0 ns per division and the Y-gain at 10 V per division.



What is the frequency of the signal?

A 2.0×10^{-8} Hz **C** 5.0×10^{7} Hz

B 2.5×10^{-2} Hz **D** 3.1×10^{8} Hz

87 **9702/13/M/J/14/Q6**

A digital caliper is used to measure the 28.50 mm width of a plastic ruler. The digital caliper reads to the nearest 0.01 mm.

What is the correct way to record this reading?

A 0.02850 ± 0.01 m **C** $(2.850 \pm 0.001) \times 10^{-2}$ m

B 0.0285 ± 0.001 m **D** $(2.850 \pm 0.001) \times 10^{-3}$ m

88 **9702/13/M/J/14/Q7**

An experiment is performed to measure the acceleration of free fall *g*. A body falls between two fixed points. The four measurements shown below are taken.

Which measurement is **not** required for the calculation of *g*?

A the distance fallen by the body **C** the mass of the body

B the initial velocity of the body **D** the time taken for the body to fall

89 **9702/13/M/J/14/Q8**

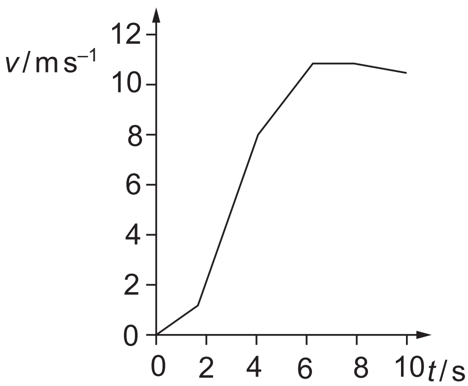
In a cathode-ray tube, an electron is accelerated uniformly in a straight line from a speed of $4 \times 10^{3}\,\mathrm{m\,s^{-1}}$ to $2 \times 10^{7}\,\mathrm{m\,s^{-1}}$ over a distance of 10 mm.

What is the acceleration of the electron?

A $2 \times 10^{3}\,\mathrm{m\,s^{-2}}$ **B** $2 \times 10^{6}\,\mathrm{m\,s^{-2}}$ **C** $2 \times 10^{13}\,\mathrm{m\,s^{-2}}$ **D** $2 \times 10^{16}\,\mathrm{m\,s^{-2}}$

90 **9702/13/M/J/14/Q9**

The graph shows how the speed *v* of a sprinter changes with time *t* during a 100 m race.



What is the best estimate of the maximum acceleration of the sprinter?

A $0.5\,\mathrm{m\,s^{-2}}$ **B** $1\,\mathrm{m\,s^{-2}}$

C $3\,\mathrm{m\,s^{-2}}$ **D** $10\,\mathrm{m\,s^{-2}}$

91 9702/11/O/N/14/Q4

A steel wire is stretched in an experiment to determine the Young modulus for steel.

The uncertainties in the measurements are given below.

measurement	uncertainty
load on wire	±2%
length of wire	±0.2%
diameter of wire	±1.5%
extension	±1%

What is the percentage uncertainty in the Young modulus?

A 1.3% **B** 1.8%

C 4.7% **D** 6.2%

92 9702/12/O/N/14/Q5

The acceleration of free fall on the Moon is one-sixth of that on Earth.

On Earth it takes time t for a stone to fall from rest a distance of 2 m.

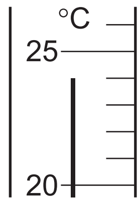
What is the time taken for a stone to fall from rest a distance of 2 m on the Moon?

A $6t$ **B** $\frac{t}{6}$ **C** $t\sqrt{6}$ **D** $\frac{t}{\sqrt{6}}$

93 9702/13/O/N/14/Q4

The diagram shows part of a thermometer.

What is the correct reading on the thermometer and the uncertainty in this reading?



	reading / °C	uncertainty in reading / °C
A	24	±1
B	24	±0.5
C	24	±0.2
D	24.0	±0.5

94 9702/13/O/N/14/Q5

The resistance R of a resistor is to be determined. The current I in the resistor and the potential difference V across it are measured.

The results, with their uncertainties, are

$$I = (2.0 \pm 0.2)\,\text{A} \qquad V = (15.0 \pm 0.5)\,\text{V}.$$

The value of R is calculated to be 7.5 Ω.

What is the uncertainty in this value for R?

A ±0.3 Ω **B** ±0.5 Ω **C** ±0.7 Ω **D** ±1 Ω

95 9702/13/O/N/14/Q6

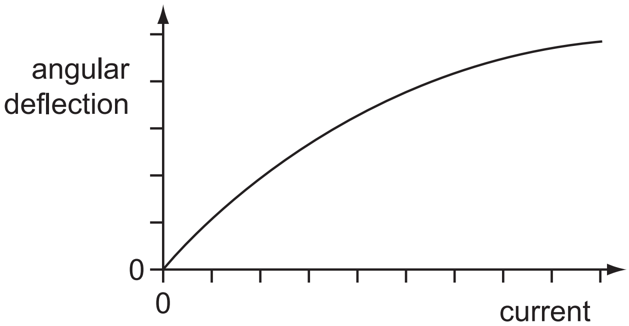
In an experiment to determine the acceleration of free fall g, a ball-bearing is held by an electromagnet. When the current to the electromagnet is switched off, a clock starts and the ball-bearing falls. After falling a distance h, the ball-bearing strikes a switch to stop the clock which measures the time t of the fall.

Which expression can be used to calculate the value of g?

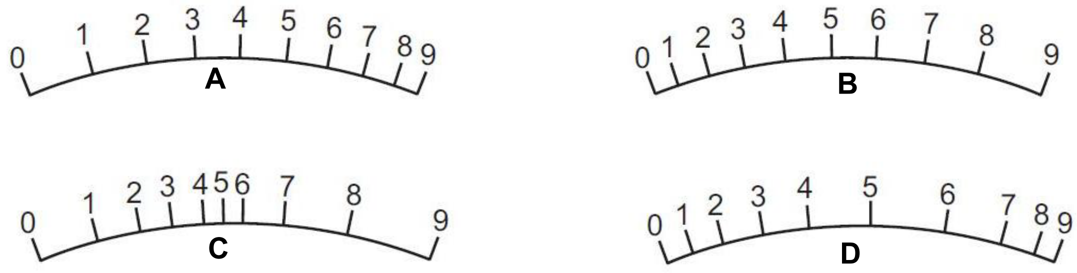
A $\frac{ht^2}{2}$ **B** $\frac{th^2}{2}$ **C** $\sqrt{\frac{2t}{h^2}}$ **D** $\frac{2h}{t^2}$

96 **9702/11/M/J/15/Q5**

The angular deflection of the needle of an ammeter varies with the current in the ammeter as shown in the graph.



Which diagram could represent the appearance of the scale on this meter?



97 **9702/11/M/J/15/Q6**

The strain energy W of a spring is determined from its spring constant k and extension x. The spring obeys Hooke's law and the value of W is calculated using the equation shown. $W = \frac{1}{2}kx^2$

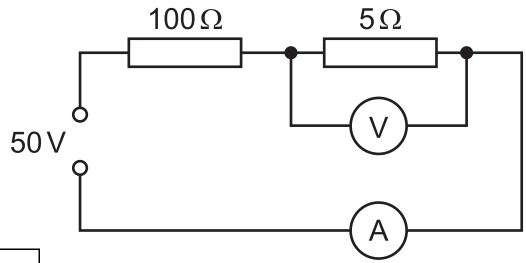
The spring constant is $100 \pm 2\,\text{N m}^{-1}$ and the extension is $0.050 \pm 0.002\,\text{m}$.

What is the percentage uncertainty in the calculated value of W?

A 6% **B** 10% **C** 16% **D** 32%

98 **9702/12/M/J/15/Q5**

A power supply of electromotive force (e.m.f.) 50 V and negligible internal resistance is connected in series with resistors of resistance 100 Ω and 5 Ω, as shown.



A voltmeter measures the potential difference (p.d.) across the 5 Ω resistor and an ammeter measures the current in the circuit.

What are suitable ranges for the ammeter and for the voltmeter?

	ammeter range / A	voltmeter range / V
A	0 – 0.1	0 – 1
B	0 – 0.1	0 – 3
C	0 – 1.0	0 – 1
D	0 – 1.0	0 – 3

99 **9702/12/M/J/15/Q6**

A single sheet of aluminium foil is folded twice to produce a stack of four sheets. The total thickness of the stack of sheets is measured to be (0.80 ± 0.02) mm. This measurement is made using a digital caliper with a zero error of (−0.20 ± 0.02) mm.

What is the percentage uncertainty in the calculated thickness of a single sheet?

A 1.0% **B** 2.0% **C** 4.0% **D** 6.7%

100 **9702/12/M/J/15/Q7**

In an experiment to determine the acceleration of free fall *g*, a ball bearing is held by an electromagnet. When the current to the electromagnet is switched off, a clock starts and the ball bearing falls. After falling a distance *h*, the ball bearing strikes a switch to stop the clock which measures the time *t* of the fall.

If systematic errors cause *t* and *h* to be measured incorrectly, which error **must** cause *g* to appear greater than $9.81\,m\,s^{-2}$?

A *h* measured as being **smaller** than it actually is and *t* is measured correctly

B *h* measured as being **smaller** than it actually is and *t* measured as being **larger** than it actually is

C *h* measured as being **larger** than it actually is and *t* measured as being **larger** than it actually is

D *h* is measured correctly and *t* measured as being **smaller** than it actually is

101 **9702/13/M/J/15/Q4**

A student is given a reel of wire of diameter less than 0.2 mm and is asked to find the density of the metal.

Which pair of instruments would be most suitable for finding the **volume** of the wire?

A balance and micrometer

B metre rule and micrometer

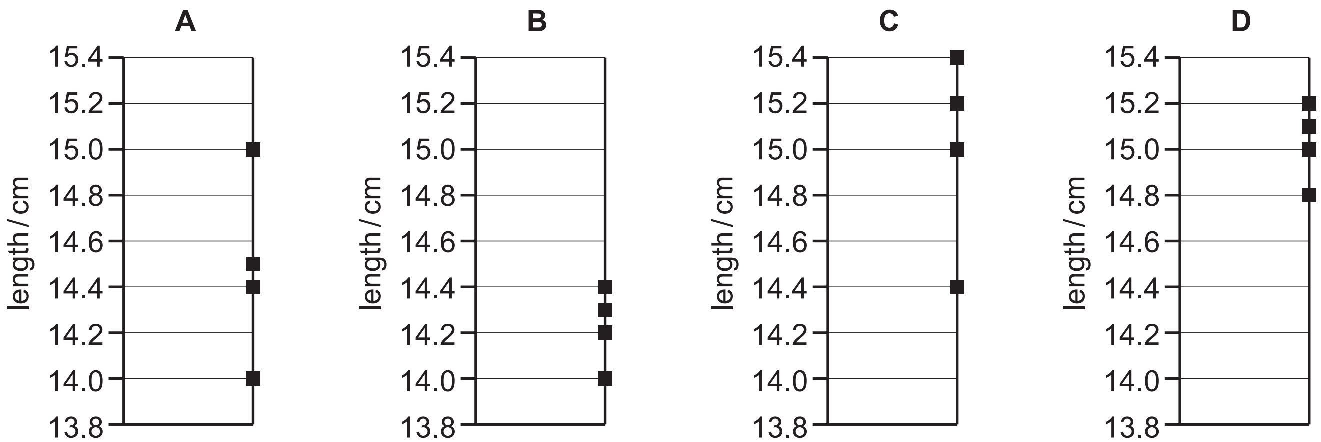
C metre rule and vernier calipers

D micrometer and vernier calipers

102 **9702/13/M/J/15/Q5**

Four different students use a ruler to measure the length of a 15.0 cm pencil. Their measurements are recorded on four different charts.

Which chart shows measurements that are precise but **not** accurate?



103 9702/11/M/J/17/Q3

The speed v of a liquid leaving a tube depends on the change in pressure ΔP and the density ρ of the liquid. The speed is given by the equation

$$v = k\left(\frac{\Delta P}{\rho}\right)^n$$

where k is a constant that has no units.

What is the value of n?

A $\frac{1}{2}$ **B** 1 **C** $\frac{3}{2}$ **D** 2

104 9702/11/M/J/17/Q9

A student attempts to find the density ρ of aluminium by taking measurements of a rectangular sheet.

mass $m = 51.6 \pm 0.1\,\text{g}$

length $l = 100.0 \pm 0.1\,\text{cm}$

width $w = 10.0 \pm 0.1\,\text{cm}$

thickness $t = 0.20 \pm 0.01\,\text{mm}$

He uses the equation $\rho = \frac{m}{wlt}$ to calculate the density.

What is the calculated value of density with its uncertainty?

A $0.26 \pm 0.01\,\text{g}\,\text{cm}^{-3}$

B $0.26 \pm 0.02\,\text{g}\,\text{cm}^{-3}$

C $2.6 \pm 0.1\,\text{g}\,\text{cm}^{-3}$

D $2.6 \pm 0.2\,\text{g}\,\text{cm}^{-3}$

105 9702/11/M/J/17/Q1

A student creates a table to show reasonable estimates of some physical quantities.

Which row is **not** a reasonable estimate?

	quantity	value
A	current in a fan heater	12 A
B	mass of an adult person	70 kg
C	speed of an Olympic sprint runner	$10\,\text{m}\,\text{s}^{-1}$
D	water pressure at the bottom of a garden pond	$10^6\,\text{Pa}$

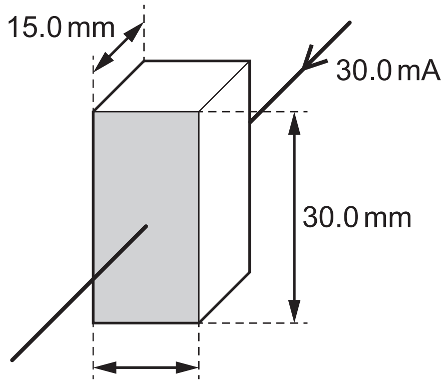
106 9702/12/M/J/17/Q1

What is the approximate average speed of a winning female Olympic athlete running a 100 m race?

A $6\,\text{m}\,\text{s}^{-1}$ **B** $9\,\text{m}\,\text{s}^{-1}$ **C** $12\,\text{m}\,\text{s}^{-1}$ **D** $15\,\text{m}\,\text{s}^{-1}$

107 9702/12/M/J/17/Q4

The current in a block of semiconductor is 30.0 mA when there is a potential difference (p.d.) of 10.0 V across it. The dimensions of the block and the direction of the current in it are as shown.



The electrical meters used are accurate to ± 0.1 mA and ± 0.1 V. The dimensions of the block are accurate to ± 0.2 mm.

What is the resistivity of the semiconductor?

A $10.0 \pm 0.2\,\Omega\,m$

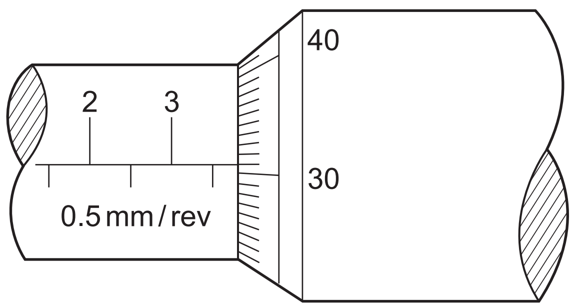
B $10.0 \pm 0.3\,\Omega\,m$

C $10.0 \pm 0.5\,\Omega\,m$

D $10.0 \pm 0.8\,\Omega\,m$

108 9702/12/M/J/17/Q5

The diameter of a cylindrical metal rod is measured using a micrometer screw gauge.
The diagram below shows an enlargement of the scale on the micrometer screw gauge when taking the measurement.



What is the cross-sectional area of the rod?

A 3.81 mm^2 **B** 11.4 mm^2 **C** 22.8 mm^2 **D** 45.6 mm^2

109 9702/11/O/N/17/Q4

What is a typical value of the wavelength of a microwave travelling in a vacuum?

A 3 000 000 pm **B** 30 nm **C** 30 000 μm **D** 3000 mm

110 **9702/11/O/N/17/Q5**

A double-slit interference experiment is used to determine the wavelength of light from a monochromatic source.

The following measurements are used.

slit separation a = 0.50 ± 0.02 mm

fringe separation x = 1.7 ± 0.1 mm

distance between slits and screen D = 2.000 ± 0.002 m

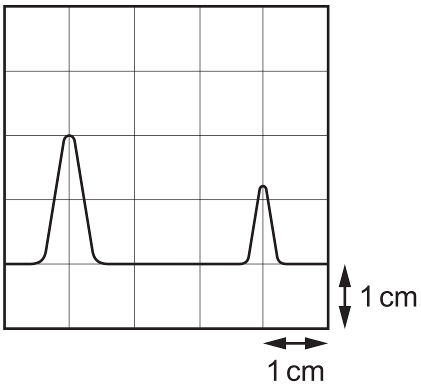
What is the percentage uncertainty in the calculated wavelength?

A 0.1% **B** 1% **C** 6% **D** 10%

111 **9702/12/O/N/17/Q5**

A transmitter emits a pulse of electromagnetic waves towards a reflector. The pulse is reflected and returns to the transmitter.

A detector is located at the transmitter. The emitted pulse and the reflected pulse are displayed on a cathode-ray oscilloscope (c.r.o.) as shown.



The pulse takes 6.3 μs to travel from the transmitter to the reflector.
What is the time-base setting of the c.r.o.?

A 2.1 $\mu s\,cm^{-1}$ **B** 3.2 $\mu s\,cm^{-1}$ **C** 4.2 $\mu s\,cm^{-1}$ **D** 6.3 $\mu s\,cm^{-1}$

112 **9702/12/O/N/17/Q6**

A hot-air balloon is moving vertically upwards with a constant speed of 3.00 $m\,s^{-1}$. A sandbag is dropped from the balloon. It takes 5.00 s for the sandbag to fall to the ground.

What was the height of the balloon when the sandbag was released?

A 29 m **B** 108 m **C** 123 m **D** 138 m

113 **9702/13/O/N/17/Q4**

A school has a piece of aluminium that it uses for radioactivity experiments. Its thickness is marked as 3.2 mm. A student decides to check this value. He has vernier calipers which give measurements to 0.1 mm and a micrometer which gives measurements to 0.01 mm.

Which statement **must** be correct?

A The micrometer gives a more accurate measurement.

B The micrometer gives a more precise measurement.

C The vernier calipers give a more accurate measurement.

D The vernier calipers give a more precise measurement.

114 **9702/13/O/N/17/Q5**

Four possible sources of error in a series of measurements are listed.

1 an analogue meter whose scale is read from different angles

2 a meter which always measures 5% too high

3 a meter with a needle that is not frictionless, so the needle sometimes sticks slightly

4 a meter with a zero error

Which errors are random and which are systematic?

	random error	systematic error
A	1 and 2	3 and 4
B	1 and 3	2 and 4
C	2 and 4	1 and 3
D	3 and 4	1 and 2

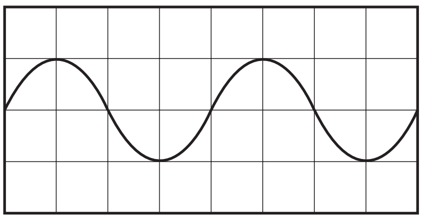
115 **9702/13/O/N/17/Q24**

An electrical signal is displayed on a cathode-ray oscilloscope (c.r.o.).

The time-base setting is $50\,\text{ms div}^{-1}$.

The Y-gain setting is $2\,\text{V div}^{-1}$.

What is the amplitude of the signal?

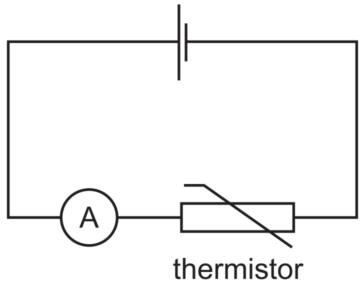


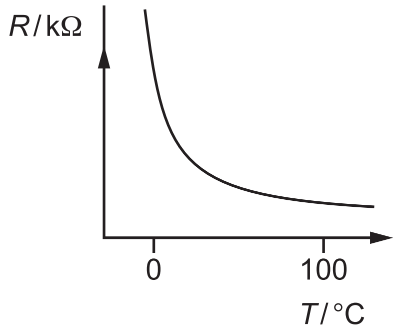
A 2 V **B** 4 V

C 5 V **D** 10 V

116 9702/11/M/J/18/Q4

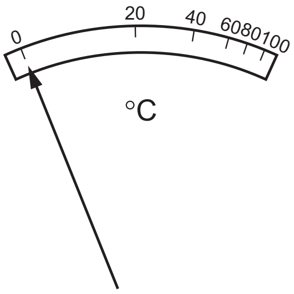
In the circuit shown, an analogue ammeter is to be recalibrated as a thermometer. The ammeter is connected in series with a thermistor. The thermistor is a component with a resistance that varies with temperature. The graph shows how the resistance R of the thermistor changes with temperature T.



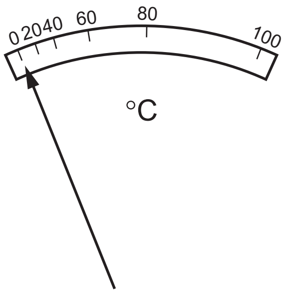


Which diagram could represent the temperature scale on the ammeter?

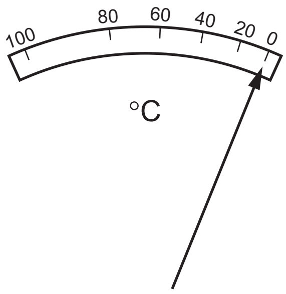
A



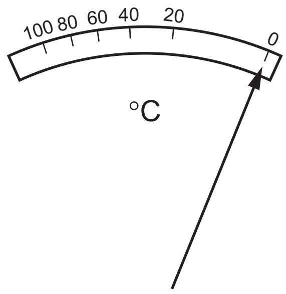
B



C



D



117 9702/11/M/J/18/Q5

The sides of a cube are measured with calipers.

The measured length of each side is (30.0 ± 0.1) mm.

The measurements are used to calculate the volume of the cube.

What is the percentage uncertainty in the calculated value of the volume?

A 0.01% **B** 0.3% **C** 1% **D** 3%

118 9702/12/M/J/18/Q3

A student measures the current through a resistor and the potential difference (p.d.) across it. There is a 4% uncertainty in the current reading and a 1% uncertainty in the p.d. reading. The student calculates the resistance of the resistor.

What is the percentage uncertainty in the calculated resistance?

A 0.25% **B** 3% **C** 4% **D** 5%

119 9702/12/M/J/18/Q4

A student applies a potential difference V of (4.0 ± 0.1) V across a resistor of resistance R of $(10.0 \pm 0.3)\,\Omega$ for a time t of (50 ± 1) s.

The student calculates the energy E dissipated using the equation below.

$$E = \frac{V^2 t}{R} = \frac{4.0^2 \times 50}{10.0} = 80\,\text{J}$$

What is the absolute uncertainty in the calculated energy value?

A 1.5 J **B** 3 J **C** 6 J **D** 8 J

120 9702/13/M/J/18/Q4

What will reduce the systematic errors when taking a measurement?

A adjusting the needle on a voltmeter so that it reads zero when there is no potential difference across it

B measuring the diameter of a wire at different points and taking the average

C reducing the parallax effects by using a marker and a mirror when measuring the amplitude of oscillation of a pendulum

D timing 20 oscillations, rather than a single oscillation, when finding the period of a pendulum

121 9702/13/M/J/18/Q5

In an experiment to determine the Young modulus E of the material of a wire, the measurements taken are shown.

mass hung on end of wire	$m = 2.300 \pm 0.002\,\text{kg}$
original length of wire	$l = 2.864 \pm 0.005\,\text{m}$
diameter of wire	$d = 0.82 \pm 0.01\,\text{mm}$
extension of wire	$e = 7.6 \pm 0.2\,\text{mm}$

The Young modulus is calculated using $E = \dfrac{4mgl}{\pi d^2 e}$

where g is the acceleration of free fall.

The calculated value of E is $1.61 \times 10^{10}\,\text{N}\,\text{m}^{-2}$.

How should the calculated value of E and its uncertainty be expressed?

A $(1.61 \pm 0.04) \times 10^{10}\,\text{N}\,\text{m}^{-2}$

B $(1.61 \pm 0.05) \times 10^{10}\,\text{N}\,\text{m}^{-2}$

C $(1.61 \pm 0.07) \times 10^{10}\,\text{N}\,\text{m}^{-2}$

D $(1.61 \pm 0.09) \times 10^{10}\,\text{N}\,\text{m}^{-2}$

122 9702/12/F/M/18/Q4

The density of paper is $800\,\text{kg}\,\text{m}^{-3}$. A typical sheet of paper has a width of 210 mm and a length of 300 mm.

The thickness of a pack of 500 sheets of paper is 50 mm.

What is the mass of a single sheet of paper?

A 0.5 g **B** 5 g **C** 50 g **D** 500 g

123 9702/12/F/M/18/Q5

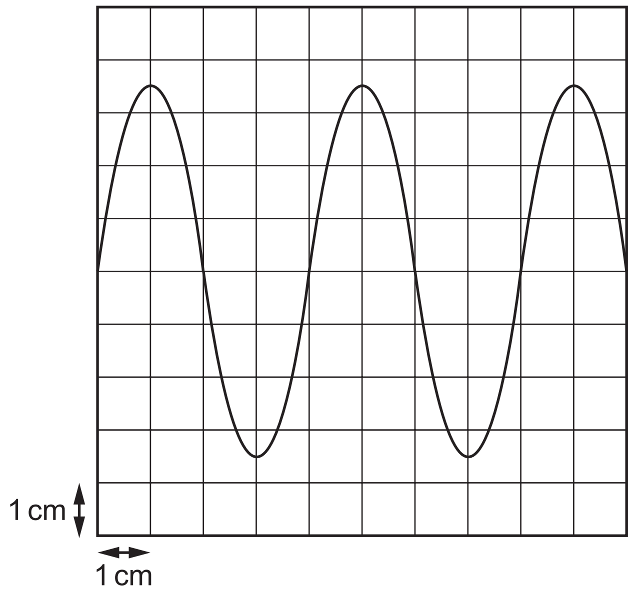
A person calculates the potential difference across a wire by using the measurements shown.

Which measured quantity has the greatest contribution to the percentage uncertainty in the calculated potential difference?

	quantity	value	uncertainty
A	current / A	5.0	± 0.5
B	diameter of wire / mm	0.8	± 0.1
C	length of wire / m	150	± 5
D	resistivity of metal in wire / Ω m	1.6×10^{-8}	$\pm 0.2 \times 10^{-8}$

124 9702/12/F/M/18/Q6

A cathode-ray oscilloscope (c.r.o.) is connected to an alternating voltage. The following trace is produced on the screen.



The oscilloscope time-base setting is 0.5 ms cm^{-1} and the Y-plate sensitivity is 2 V cm^{-1}.

Which statement about the alternating voltage is correct?

A The amplitude is 3.5 cm.

B The frequency is 0.5 kHz.

C The period is 1 ms.

D The wavelength is 4 cm.

125 **9702/11/O/N/18/Q4**

A micrometer screw gauge is used to measure the diameter of a copper wire.

The reading with the wire in position is shown in diagram 1. The wire is removed and the jaws of the micrometer are closed. The new reading is shown in diagram 2.

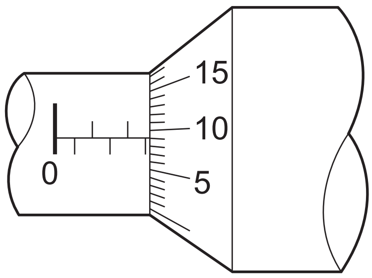


diagram 1

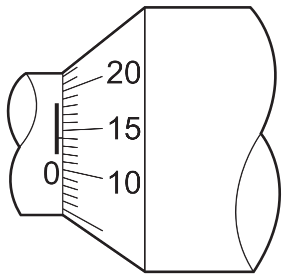


diagram 2

What is the diameter of the wire?

A 1.90 mm **B** 2.45 mm **C** 2.59 mm **D** 2.73 mm

126 **9702/11/O/N/18/Q5**

A digital meter has an accuracy of ±1%.

The meter is used to measure the current in an electrical circuit.

The reading on the meter varies between 3.04 A and 3.08 A.

What is the value of the current, with its uncertainty?

A (3.06 ± 0.02) A **C** (3.06 ± 0.05) A

B (3.06 ± 0.04) A **D** (3.06 ± 0.07) A

127 **9702/12/O/N/18/Q1**

A car is travelling at a speed of 20 m s^{-1}. The table contains values for the kinetic energy and the momentum of the car.

Which values are reasonable estimates?

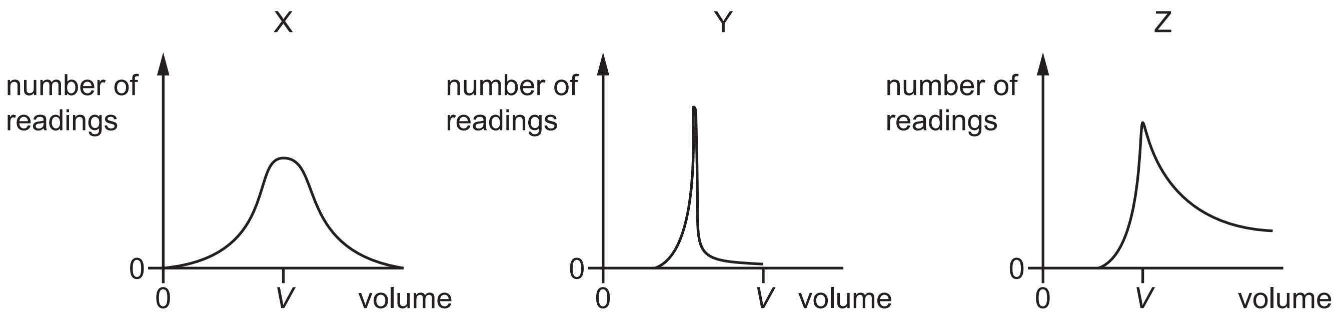
	kinetic energy / J	momentum / kg m s^{-1}
A	3×10^5	3×10^4
B	3×10^5	5×10^6
C	2×10^7	3×10^4
D	2×10^7	5×10^6

128 9702/12/O/N/18/Q5

Students take readings of the volume of a liquid using three different pieces of measuring equipment X, Y and Z.

The true value of the volume of the liquid is V.

The students' results are shown.



How many pieces of equipment are precise and how many are accurate?

	number of precise pieces of equipment	number of accurate pieces of equipment
A	1	1
B	1	2
C	2	1
D	2	2

129 9702/13/O/N/18/Q1

Which statement is **not** a reasonable estimate?

A Atmospheric pressure at sea level is about 1×10^5 Pa.

B Light takes 5×10^2 s to reach us from the Sun.

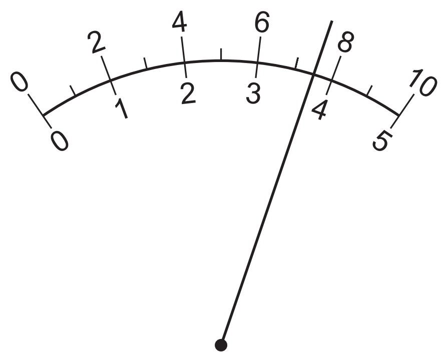
C The frequency of ultraviolet light is 3×10^{12} Hz.

D The lifespan of a man is about 2×10^9 s.

130 **9702/13/O/N/18/Q4**

An ammeter is calibrated so that it shows a full-scale deflection when it measures a current of 2.0 A.

The diagram shows the display of this ammeter when it is measuring a current.



Which current is the ammeter measuring?

A 0.75 A **B** 1.5 A **C** 3.8 A **D** 7.5 A

131 **9702/13/O/N/18/Q5**

The width of a table is measured as (50.3 ± 0.1) cm. Its length is measured as (1.40 ± 0.01) m.

What is the area of the table and its absolute uncertainty?

A $(0.7 \pm 0.1)\,m^2$

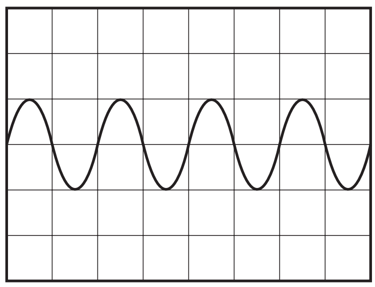
B $(0.704 \pm 0.006)\,m^2$

C $(0.704 \pm 0.011)\,m^2$

D $(70.4 \pm 0.6)\,m^2$

132 **9702/11/M/J/19/Q4**

A whale produces sound waves of frequency 5 Hz. The waves are detected by a microphone and displayed on an oscilloscope.



What is the time-base setting on the oscilloscope?

A $0.1\,ms\,div^{-1}$ **B** $1\,ms\,div^{-1}$ **C** $10\,ms\,div^{-1}$ **D** $100\,ms\,div^{-1}$

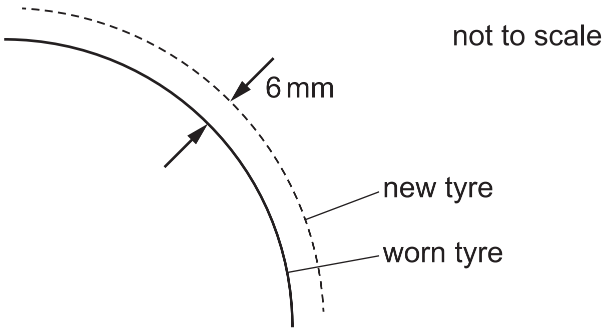
133 9702/11/M/J/19/Q5

The speed shown on a car's speedometer is proportional to the rate of rotation of the tyres.

The variation of the diameter of a tyre as it wears introduces an error in the speed shown on the speedometer.

A car has new tyres of diameter 600 mm. The speedometer is accurately calibrated for this diameter.

The tyres wear as shown, with 6 mm of material being removed from the outer surface.



What is the error in the speed shown on the speedometer after this wear has taken place?

A The speed shown is too high by 1%.

B The speed shown is too high by 2%.

C The speed shown is too low by 1%.

D The speed shown is too low by 2%.

134 9702/12/M/J/19/Q5

A student wishes to determine the density ρ of lead. She measures the mass and diameter of a small sphere of lead:

$$\text{mass} = (0.506 \pm 0.005)\,\text{g}$$

$$\text{diameter} = (2.20 \pm 0.02)\,\text{mm.}$$

What is the best estimate of the percentage uncertainty in her calculated value of ρ?

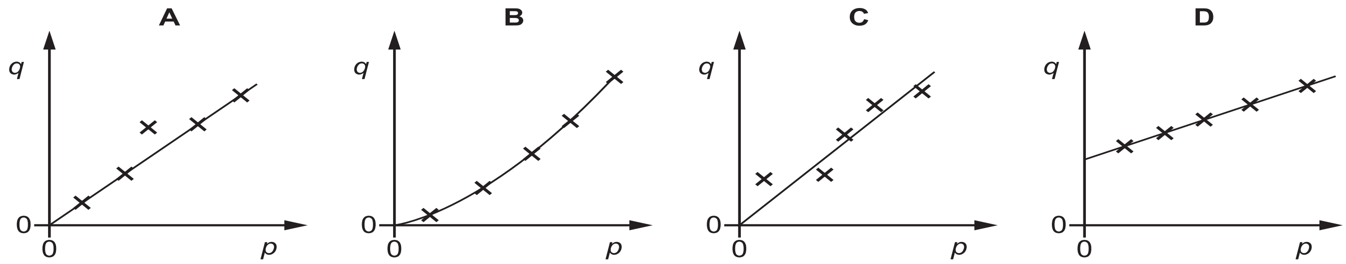
A 1.7% **B** 1.9% **C** 2.8% **D** 3.7%

135 9702/12/M/J/19/Q6

Two quantities p and q are directly proportional to each other.

Experimental results are taken and plotted in a graph of q against p.

Which graph shows there were random errors in the measurements of p and q?



136 9702/12/M/J/19/Q7

A man of mass 75.2 kg uses a set of weighing scales to measure his mass three times. He obtains the following readings.

	mass / kg
reading 1	80.2
reading 2	80.1
reading 3	80.2

Which statement best describes the precision and accuracy of the weighing scales?

A not precise to ± 0.1 kg and accurate to ± 0.1 kg

B not precise to ± 0.1 kg and not accurate to ± 0.1 kg

C precise to ± 0.1 kg and accurate to ± 0.1 kg

D precise to ± 0.1 kg and not accurate to ± 0.1 kg

137 9702/13/M/J/19/Q4

What is the approximate kinetic energy of an Olympic athlete when running at maximum speed during a 100 m race?

A 400 J **B** 4000 J **C** 40 000 J **D** 400 000 J

138 9702/13/M/J/19/Q6

A micrometer screw gauge is used to measure the diameter of a small uniform steel sphere. The micrometer reading is 5.00 mm ± 0.01 mm.

What will be the percentage uncertainty in a calculation of the volume of the sphere, using these values?

A 0.2% **B** 0.4% **C** 0.6% **D** 1.2%

139 **9702/11/O/N/19/Q1**

For which quantity is the magnitude a reasonable estimate?

A frequency of a radio wave 500 pHz

B mass of an atom 500 μg

C the Young modulus of a metal 500 kPa

D wavelength of green light 500 nm

140 **9702/11/O/N/19/Q2**

The speed of a wave in deep water depends on its wavelength *L* and the acceleration of free fall *g*.

What is a possible equation for the speed *v* of the wave?

A $v = \sqrt{\left(\frac{gL}{2\pi}\right)}$ **B** $v = \frac{gL}{4\pi^2}$ **C** $v = 2\pi\sqrt{\left(\frac{g}{L}\right)}$ **D** $v = \frac{2\pi g}{L}$

141 **9702/11/O/N/19/Q4**

A student intends to measure accurately the diameter of a wire (known to be approximately 1 mm) and the internal diameter of a pipe (known to be approximately 2 cm).

What are the most appropriate instruments for the student to use to make these measurements?

	wire	pipe
A	calipers	calipers
B	calipers	micrometer
C	micrometer	calipers
D	micrometer	micrometer

142 **9702/11/O/N/19/Q5**

The power *P* dissipated in a resistor of resistance *R* is calculated using the expression

$$P = \frac{V^2}{R}$$

where *V* is the potential difference (p.d.) across the resistor. The percentage uncertainty in *V* is 5% and in *R* is 2%.

What is the percentage uncertainty in *P*?

A 3% **B** 7% **C** 8% **D** 12%

143 **9702/12/O/N/19/Q1**

A cyclist has a speed of $5\,m\,s^{-1}$ and a small car has a speed of $12\,m\,s^{-1}$.

Which statement does **not** give a reasonable estimate?

A The kinetic energy of the cyclist is 1×10^{3} J.

B The kinetic energy of the car is 7×10^{4} J.

C The momentum of the cyclist is $4 \times 10^{2}\,kg\,m\,s^{-1}$.

D The momentum of the car is $2 \times 10^{5}\,kg\,m\,s^{-1}$.

144 **9702/12/O/N/19/Q4**

A micrometer is used to measure the 28.50 mm width of a plastic ruler. The micrometer reads to the nearest 0.01 mm.

What is the correct way to record this reading?

A 0.02850 ± 0.01 m

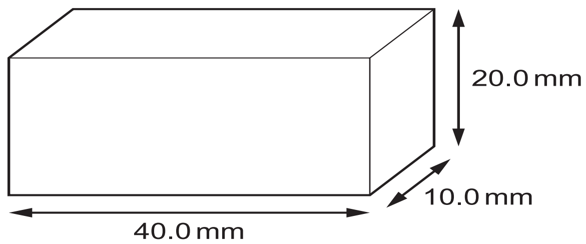
B 0.0285 ± 0.001 m

C $(2.850 \pm 0.001) \times 10^{-2}$ m

D $(2.850 \pm 0.001) \times 10^{-3}$ m

145 **9702/12/O/N/19/Q5**

The sides of a wooden block are measured with calipers. The lengths of the sides are measured as 20.0 mm, 40.0 mm and 10.0 mm.



The calipers can measure with an absolute uncertainty of ± 0.1 mm.

What is the percentage uncertainty in the calculated volume of the block?

A 0.3% **B** 1.8% **C** 3.8% **D** 30%

146 **9702/13/O/N/19/Q4**

What could reduce systematic errors?

A averaging a large number of measurements

B careful calibration of measuring instruments

C reducing the sample size

D repeating measurements

147 9702/13/O/N/19/Q5

The power loss P in a resistor is calculated using the formula $P = \frac{V^2}{R}$.

The percentage uncertainty in the potential difference V is 3% and the percentage uncertainty in the resistance R is 2%.

What is the percentage uncertainty in P?

A 4% **B** 7% **C** 8% **D** 11%

148. 9702/12/F/M/20/Q5

A micrometer is used to measure the diameters of two cylinders.

diameter of first cylinder = (12.78 ± 0.02) mm

diameter of second cylinder = (16.24 ± 0.03) mm

The difference in the diameters is calculated.

What is the uncertainty in this difference?

A 0.01 mm **B** 0.02 mm **C** 0.03 mm **D** 0.05 mm

149. 9702/11/M/J/20/Q5

A measurement is taken correctly but with a ruler at a significantly higher temperature than that at which the ruler was calibrated. The higher temperature causes the ruler to expand.

What describes the effect on the measurement caused by the higher temperature and how the measurement may be improved?

A The measurement will be subject to a random error. The measurement can be made more accurate by taking the average of several repeated measurements.

B The measurement will be subject to a random error. The measurement can be made more precise by taking the average of several repeated measurements.

C The measurement will be subject to a systematic error. The measurement can be made more accurate by taking the average of several repeated measurements.

D The measurement will be subject to a systematic error. The measurement can be made more precise by taking the average of several repeated measurements.

150. 9702/12/M/J/20/Q3

A galvanometer of resistance 5 Ω is to be used in a null method.

In order to protect the galvanometer from damage due to an excessive initial current, resistors of resistance 0.5 Ω and 1 kΩ are available.

Which arrangement would provide this protection?

A the 0.5 Ω resistor in series with the galvanometer

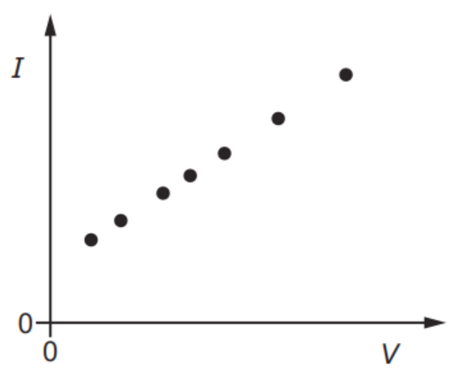
B the 0.5 Ω resistor in parallel with the galvanometer and this combination placed in series with the 1 kΩ resistor

C the 1 kΩ resistor in parallel with the galvanometer

D the 1 kΩ resistor in parallel with the galvanometer and this combination placed in series with the 0.5 Ω resistor

151. 9702/12/M/J/20/Q4

Readings are made of the current I for different voltages V across a fixed resistor. The results are plotted on a graph to show the variation of I with V.



What is the best description of the errors in the readings?

A both systematic and random

B neither systematic nor random

C random only

D systematic only

152. 9702/13/M/J/20/O4

A circuit is set up in order to determine the resistance of a 12 V, 1.2 W lamp when operating normally. An analogue ammeter and an analogue voltmeter are used.

Which ranges for the meters would be most suitable?

	ammeter range / A	voltmeter range / V
A	0–0.5	0–20
B	0–0.5	0–100
C	0–10	0–20
D	0–10	0–100

153. 9702/13/M/J/20/Q5

Two liquid-in-glass thermometers in a well-mixed liquid are individually observed by 10 different students. All agree that one thermometer reads 21 °C and the other thermometer reads 23 °C.

What is a possible explanation for the difference?

A The liquid is not all at the same temperature.

B The readings are not precise.

C There is a random error affecting the readings.

D There is a systematic error affecting the readings.

154. 9702/11/O/N/20/Q5

The diameter of a spherical golf ball is measured with calipers and found to be (4.11 ± 0.01) cm.

The volume of a sphere is $V = \frac{1}{6}\pi d^3$, where d is the diameter of the sphere.

What is the volume of the golf ball?

A (36.35 ± 0.01) cm^3

B (36.35 ± 0.03) cm^3

C (36.35 ± 0.09) cm^3

D (36.4 ± 0.3) cm^3

155. 9702/12/O/N/20/Q1

A student uses the volume of a metal coin in order to determine the density of the metal.

What is **not** needed in order to determine an estimate of the volume of the coin?

A estimate of the diameter

B estimate of the mass

C estimate of the thickness

D use of the formula for the volume of a cylinder

156. 9702/12/O/N/20/Q4

A student wishes to measure a distance of about 10 cm to a precision of 0.01 cm.

Which measuring instrument should be used?

A metre rule

B micrometer

C tape measure

D vernier calipers

157. 9702/12/O/N/20/Q5

A steel ball is dropped and falls through a vertical height h. The time t taken to fall is measured using light gates.

The results are given in the table.

h	$(4.05 \pm 0.01)\,m$
t	$(0.91 \pm 0.02)\,s$

The acceleration of free fall g is calculated using the equation shown.

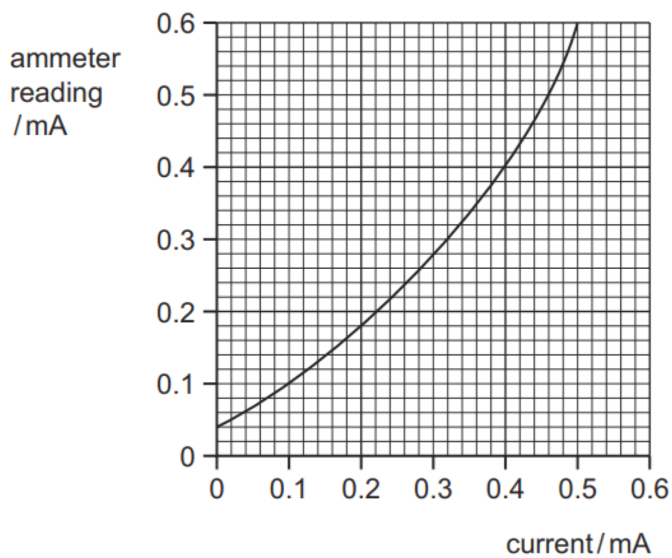
$$h = \tfrac{1}{2}gt^2$$

What is the percentage uncertainty in the value of g?

A 2.4% **B** 4.6% **C** 5.1% **D** 9.3%

158. 9702/13/O/N/20/Q4

A calibration curve is shown for an ammeter whose scale is inaccurate.



Two readings taken on the meter at different times during an experiment are 0.13 mA and 0.47 mA.

By how much did the current really increase between taking the two readings?

A 0.30 mA **B** 0.34 mA **C** 0.40 mA **D** 0.44 mA

159. 9702/13/O/N/20/Q5

A student measures the length l and the period T of oscillation of a simple pendulum. He then uses the equation shown to calculate the acceleration of free fall g.

$$T = 2\pi\sqrt{\frac{l}{g}}$$

His measurements are shown.

l	(87.3 ± 0.2) cm
T	(1.9 ± 0.05) s

What is the percentage uncertainty in his calculated value of g?

A 2.4% **B** 2.9% **C** 5.5% **D** 7.2%

160. 9702/12/F/M/21/Q4

A micrometer screw gauge is used to measure the diameter of a copper wire.

The reading with the wire in position is shown in diagram 1. The wire is removed and the jaws of the micrometer are closed. The new reading is shown in diagram 2.

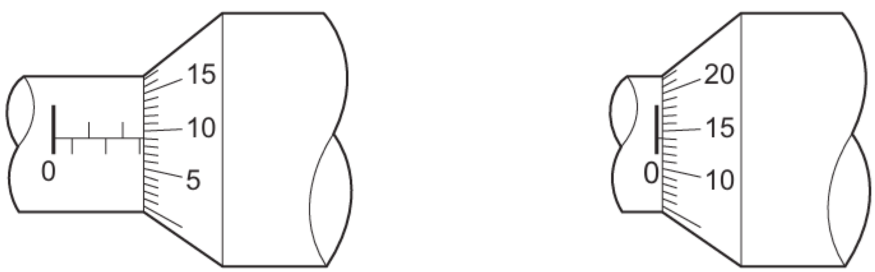


diagram 1 diagram 2

What is the diameter of the wire?

A 1.95 mm **B** 2.45 mm **C** 2.59 mm **D** 2.73 mm

161. 9702/12/F/M/21/Q5

A student measures the current and the potential difference for a resistor in a circuit.

current = (50.00 ± 0.01) mA

potential difference = (500.0 ± 0.1) mV

The measurements are used to calculate the resistance of the resistor.

What is the percentage uncertainty in the calculated resistance?

A 0.0002% **B** 0.0004% **C** 0.02% **D** 0.04%

162. 9702/11/M/J/21/Q5

A micrometer screw gauge is used to measure the diameter of a wire.

The reading on the micrometer with the jaws closed is (-0.05 ± 0.02) mm.

The reading with the wire in position between the two jaws is $(+1.03 \pm 0.02)$ mm.

What is the diameter of the wire?

A (0.98 ± 0.02) mm

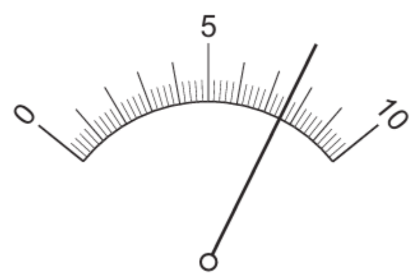
B (1.08 ± 0.02) mm

C (0.98 ± 0.04) mm

D (1.08 ± 0.04) mm

163. 9702/12/M/J/21/Q4

An analogue ammeter with a range of 0–250 mA is connected into an electrical circuit. The diagram shows the ammeter's display.



What is the reading on the ammeter?

A 76 mA **B** 165 mA **C** 183 mA **D** 190 mA

164. 9702/13/M/J/21/Q4

The diagram shows two readings on a micrometer.

reading 1 reading 2

What is the difference between the two readings?

A 10.34 mm **B** 11.84 mm **C** 12.34 mm **D** 12.84 mm

165. 9702/13/M/J/21/Q5

The diameter of a circular disc is measured as (7.0 ± 0.1) mm.

What is the area of the disc and the absolute uncertainty in the area?

	area of disc / mm^2	absolute uncertainty / mm^2
A	38.5	± 0.5
B	38	± 1
C	154	± 2
D	154	± 4

166. 9702/11/O/N/21/Q5

Four possible sources of error in a series of measurements are listed.

1 an analogue meter whose scale is read from different angles

2 a meter which always measures 5% too high

3 a meter with a needle that is not frictionless, so the needle sometimes sticks slightly

4 a meter with a zero error

Which errors are random and which are systematic?

	random error	systematic error
A	1 and 2	3 and 4
B	1 and 3	2 and 4
C	2 and 4	1 and 3
D	3 and 4	1 and 2

167. 9702/12/O/N/21/Q5

A student measures the time T for one complete oscillation of a pendulum of length l.

Her results are shown in the table.

l / m	T / s
0.420 ± 0.001	1.3 ± 0.1

She uses the formula

$$T = 2\pi\sqrt{\frac{l}{g}}$$

to calculate the acceleration of free fall g.

What is the best estimate of the percentage uncertainty in the value of g?

A 0.02% **B** 4% **C** 8% **D** 16%

168. 9702/13/O/N/21/Q5

After measuring the width of a shelf to be 305 mm, it is found that the graduations on the ruler used are 1.0% further apart than they should be.

Which type of measurement error is this and what is the true width of the shelf?

	type of error	true width / mm
A	random	302
B	random	308
C	systematic	302
D	systematic	308

169. 9702/12/F/M/22/Q3

A man of mass 75.2 kg uses a set of weighing scales to measure his mass three times. He obtains the following readings.

	mass / kg
reading 1	80.2
reading 2	80.1
reading 3	80.2

Which statement describes the precision and accuracy of the weighing scales?

A not precise to ± 0.1 kg and accurate to ± 0.1 kg

B not precise to ± 0.1 kg and not accurate to ± 0.1 kg

C precise to ± 0.1 kg and accurate to ± 0.1 kg

D precise to ± 0.1 kg and not accurate to ± 0.1 kg

170. 9702/11/M/J/22/Q3

A value for the acceleration of free fall on Earth is given as $(10 \pm 2)\,\mathrm{m\,s^{-2}}$.

Which statement is correct?

A The value is accurate but not precise.

B The value is both precise and accurate.

C The value is neither precise nor accurate.

D The value is precise but not accurate.

171. 9702/12/M/J/22/Q3

Which statement about systematic errors is **not** correct?

A A systematic error can be caused by using an incorrectly calibrated instrument.

B One particular type of systematic error can affect all the measurements by the same amount.

C The effect of a systematic error can be reduced by repeating and averaging the measurements.

D Zero error is a type of systematic error.

172. 9702/13/M/J/22/Q3

A micrometer screw gauge is used to measure the diameter of a small uniform steel sphere. The measurement of the diameter is 5.00 mm ± 0.01 mm.

What is the percentage uncertainty in the calculated volume of the sphere, using these values?

A 0.2% **B** 0.4% **C** 0.6% **D** 1.2%

173. 9702/11/O/N/22/Q3

In an experiment to determine the acceleration of free fall g, the time t taken for a ball to fall through distance s is measured. The percentage uncertainty in the measurement of s is 2%. The percentage uncertainty in the measurement of t is 3%.

The value of g is determined using the equation shown.

$$g = \frac{2s}{t^2}$$

What is the percentage uncertainty in the calculated value of g?

A 1% **B** 5% **C** 8% **D** 11%

174. 9702/12/O/N/22/Q3

A spring is suspended from a fixed point and a force is applied. The position of a pointer attached to the bottom of the spring against a vertical ruler is recorded.

Before the force is applied, the position of the pointer is (225 ± 2) mm.

After the force is applied, the position of the pointer is (250 ± 2) mm.

The extension of the spring is determined.

What is the percentage uncertainty in the extension?

A 1.6% **B** 1.8% **C** 8.0% **D** 16%

175. 9702/13/O/N/22/Q3

A digital meter is used to measure the current in an electric circuit.

The reading on the meter fluctuates (varies) between 3.04 A and 3.08 A. The readings on the meter have an accuracy of $\pm1\%$.

What is the true value of the current, with its uncertainty?

A (3.06 ± 0.02) A

B (3.06 ± 0.04) A

C (3.06 ± 0.05) A

D (3.06 ± 0.07) A

176. 9702/12/F/M/23/Q3

A hollow cylinder, which is open at both ends, has a radius of (3.0 ± 0.1) cm and a length of (15.0 ± 0.1) cm.

What is the value, with its absolute uncertainty, of the surface area of the cylinder?

A $(280 \pm 10)\,\text{cm}^2$

B $(282.7 \pm 0.2)\,\text{cm}^2$

C $(420 \pm 30)\,\text{cm}^2$

D $(424.1 \pm 0.3)\,\text{cm}^2$

177. 9702/11/M/J/23/Q3

A copper pipe has a true diameter of 42.03 mm.

A builder measures the diameter of the pipe five times using digital calipers. The measurements are shown.

diameter / mm
48.01
47.99
48.01
48.00
47.99

What describes the builder's measurements?

A accurate and precise

B accurate but not precise

C not precise and not accurate

D precise but not accurate

178. 9702/12/M/J/23/Q3

Two measurements for a solid sphere are shown.

$$\text{mass} = (32.5 \pm 0.1)\,\text{g}$$

$$\text{diameter} = (1.87 \pm 0.04)\,\text{cm}$$

These values are used to determine the density of the sphere.

What is the percentage uncertainty in the density?

A 2.4% **B** 4.6% **C** 6.1% **D** 6.7%

179. 9702/13/M/J/23/Q3

A desk has a true width of 50.0 cm.

Two students, X and Y, measure the width of the desk.

Student X uses a tape measure and records a width of (49.5 ± 0.5) cm.

Student Y uses a metre rule and records a width of (51.4 ± 0.1) cm.

Which statement about the measurement of student X is correct?

A It is less accurate and less precise than the measurement of student Y.

B It is less accurate but more precise than the measurement of student Y.

C It is more accurate and more precise than the measurement of student Y.

D It is more accurate but less precise than the measurement of student Y.

180. 9702/13/O/N/23/Q3

A set of repeated measurements is made of a fixed quantity. An average of these measurements is calculated.

What is the effect of averaging on the random error and the systematic error in the measurements?

A Random error and systematic error are both reduced.

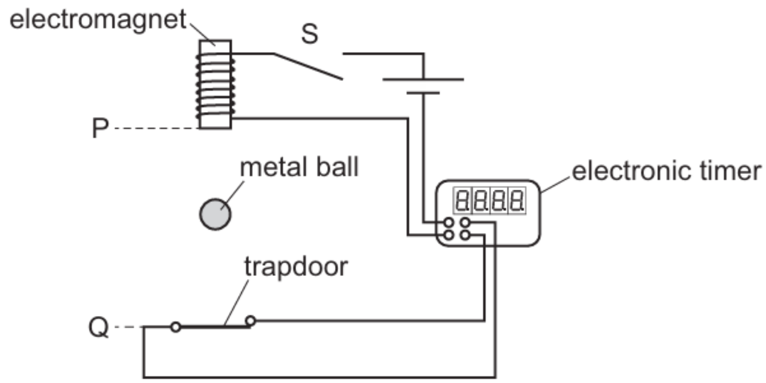
B Random error and systematic error are both unaffected.

C Random error is reduced but systematic error is unaffected.

D Random error is unaffected but systematic error is reduced.

181. 9702/11/O/N/20/Q3

A student determines the acceleration of free fall by using a small metal ball, as shown.



When switch S is opened, the ball is released from an electromagnet and an electronic timer is started. The ball then falls vertically downwards. The timer stops when the ball hits a trapdoor. The student measures the distance PQ between the electromagnet and the trapdoor. This distance and the reading on the timer are then used to calculate the acceleration of free fall.

Which statement about errors in the experiment is correct?

A The random error can be reduced by adding the diameter of the ball to the distance PQ.

B The random error can be reduced by subtracting the diameter of the ball from the distance PQ.

C The systematic error can be reduced by adding the diameter of the ball to the distance PQ.

D The systematic error can be reduced by subtracting the diameter of the ball from the distance PQ.

182. 9702/12/O/N/23/Q3

A student takes measurements to determine the constant acceleration of a model car moving from rest in a straight line. The measured values with their absolute uncertainties are shown.

quantity	measured value	uncertainty
displacement	16.5 m	± 0.1 m
time	15.0 s	± 1.0 s

The student uses the equation $s = \frac{1}{2}at^2$ to calculate the acceleration of the car.

What is the acceleration and its absolute uncertainty?

A $(0.11 \pm 0.01)\,\text{m s}^{-2}$

B $(0.11 \pm 0.02)\,\text{m s}^{-2}$

C $(0.15 \pm 0.01)\,\text{m s}^{-2}$

D $(0.15 \pm 0.02)\,\text{m s}^{-2}$